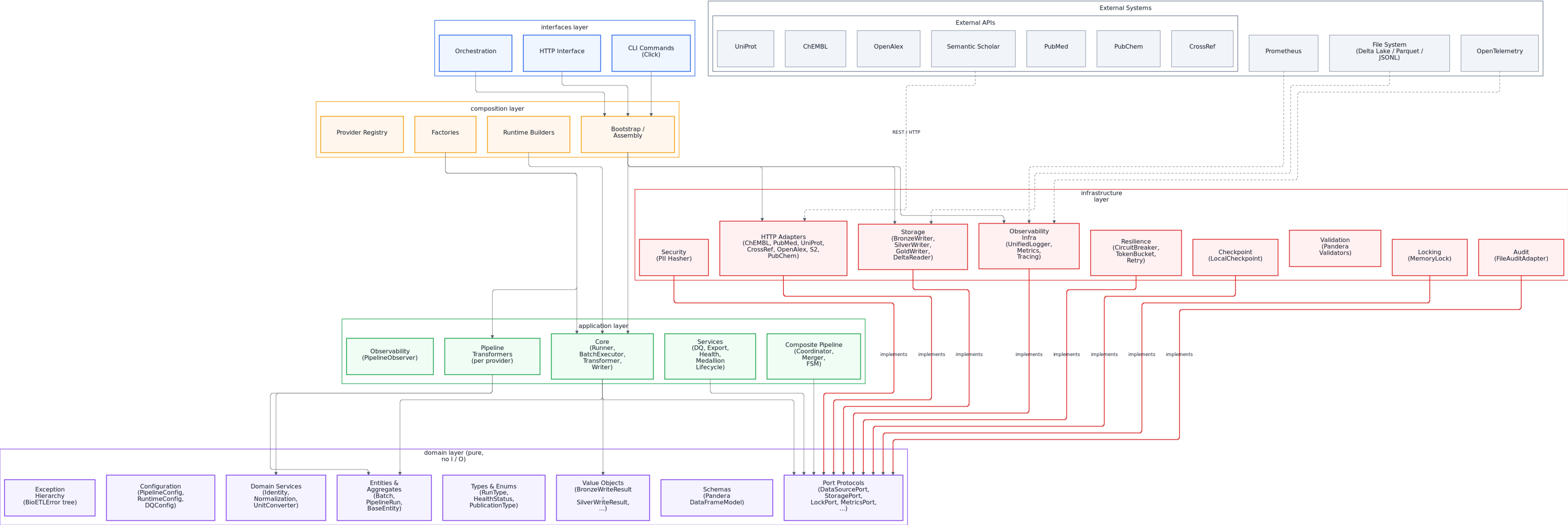
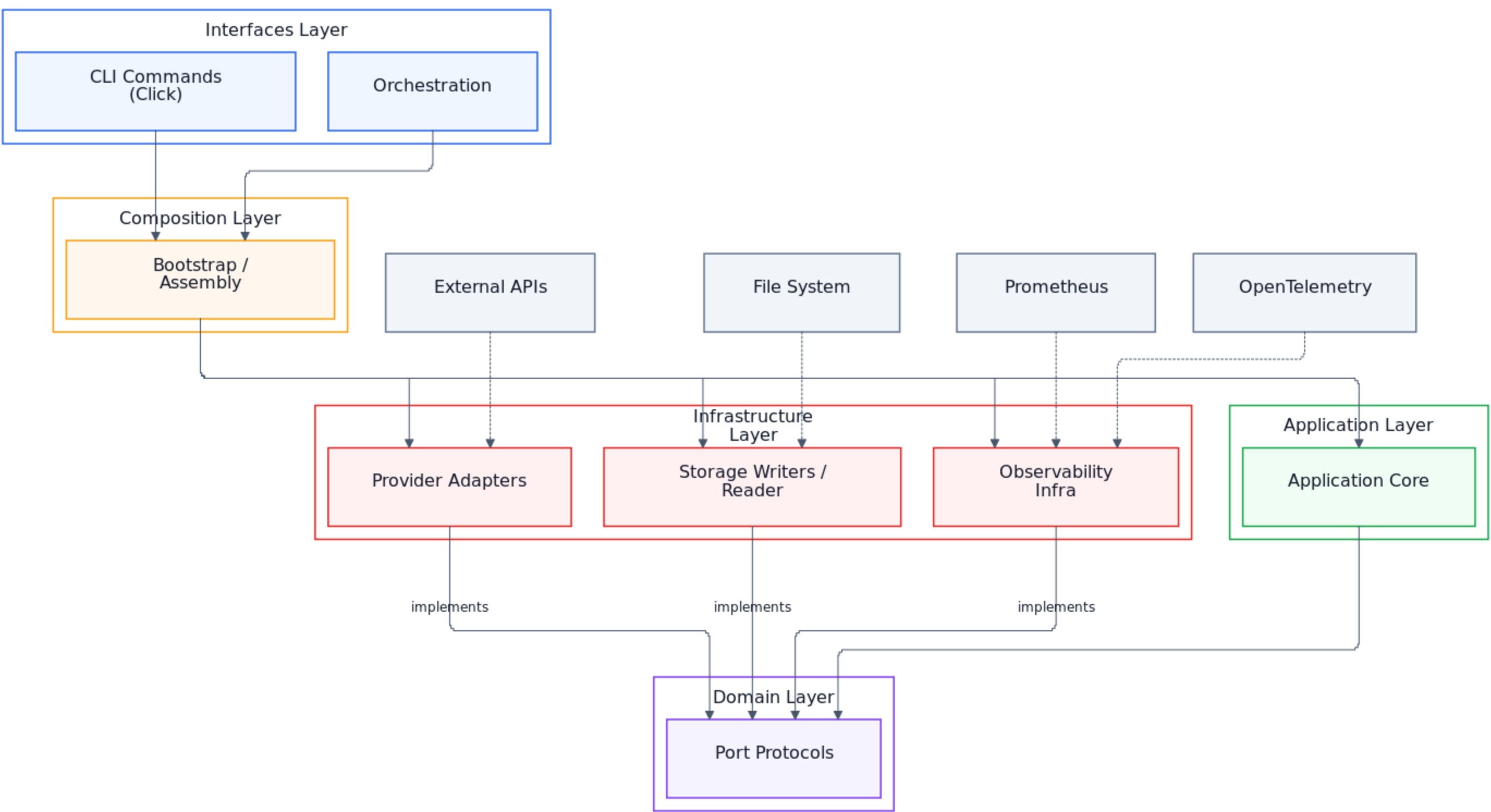
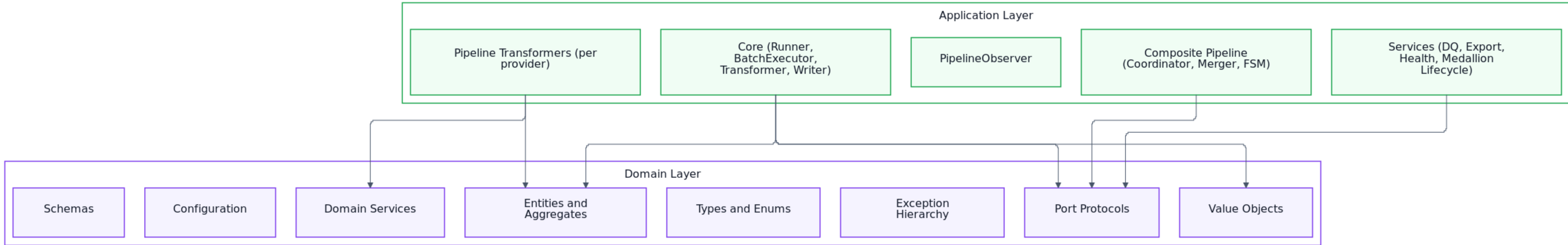
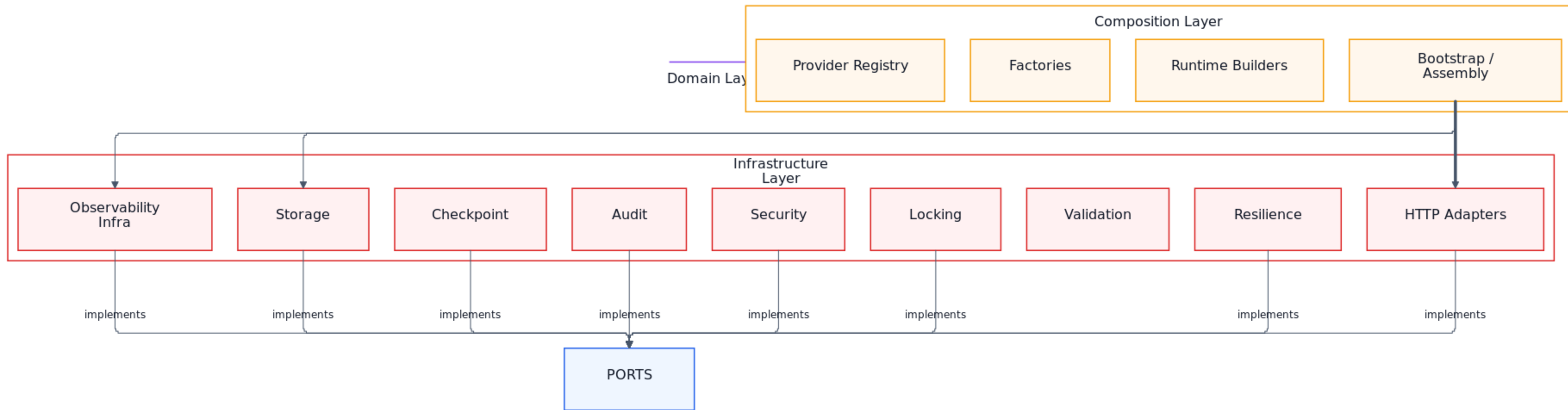
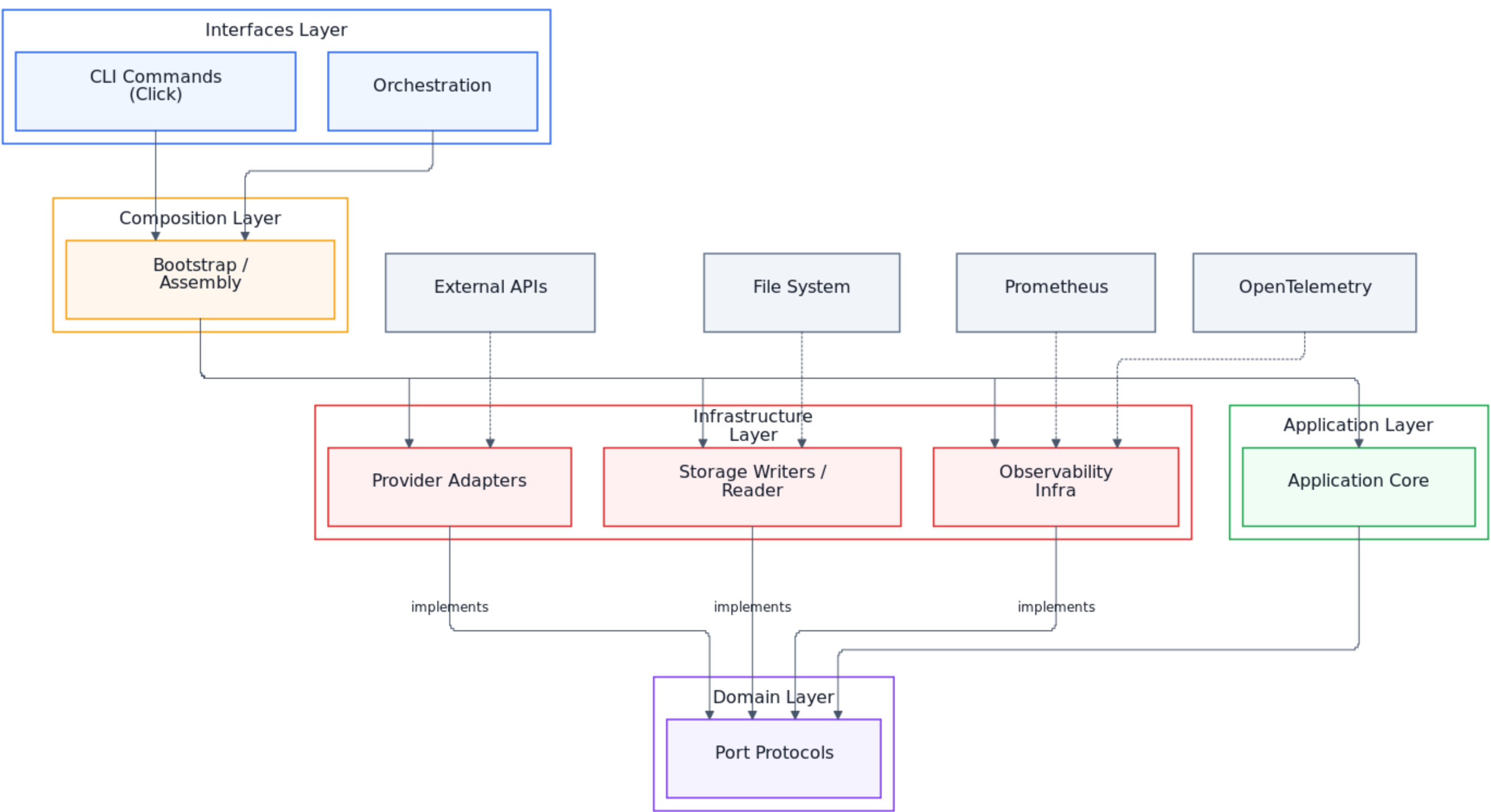


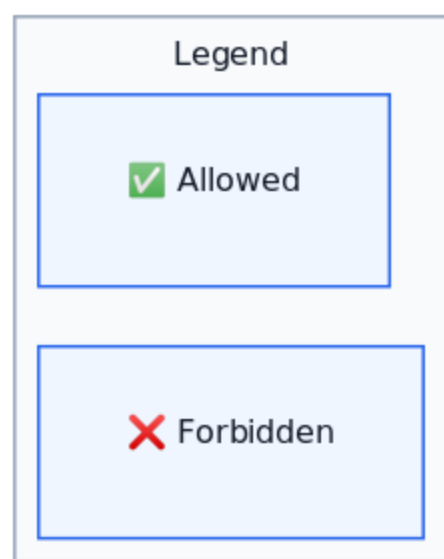
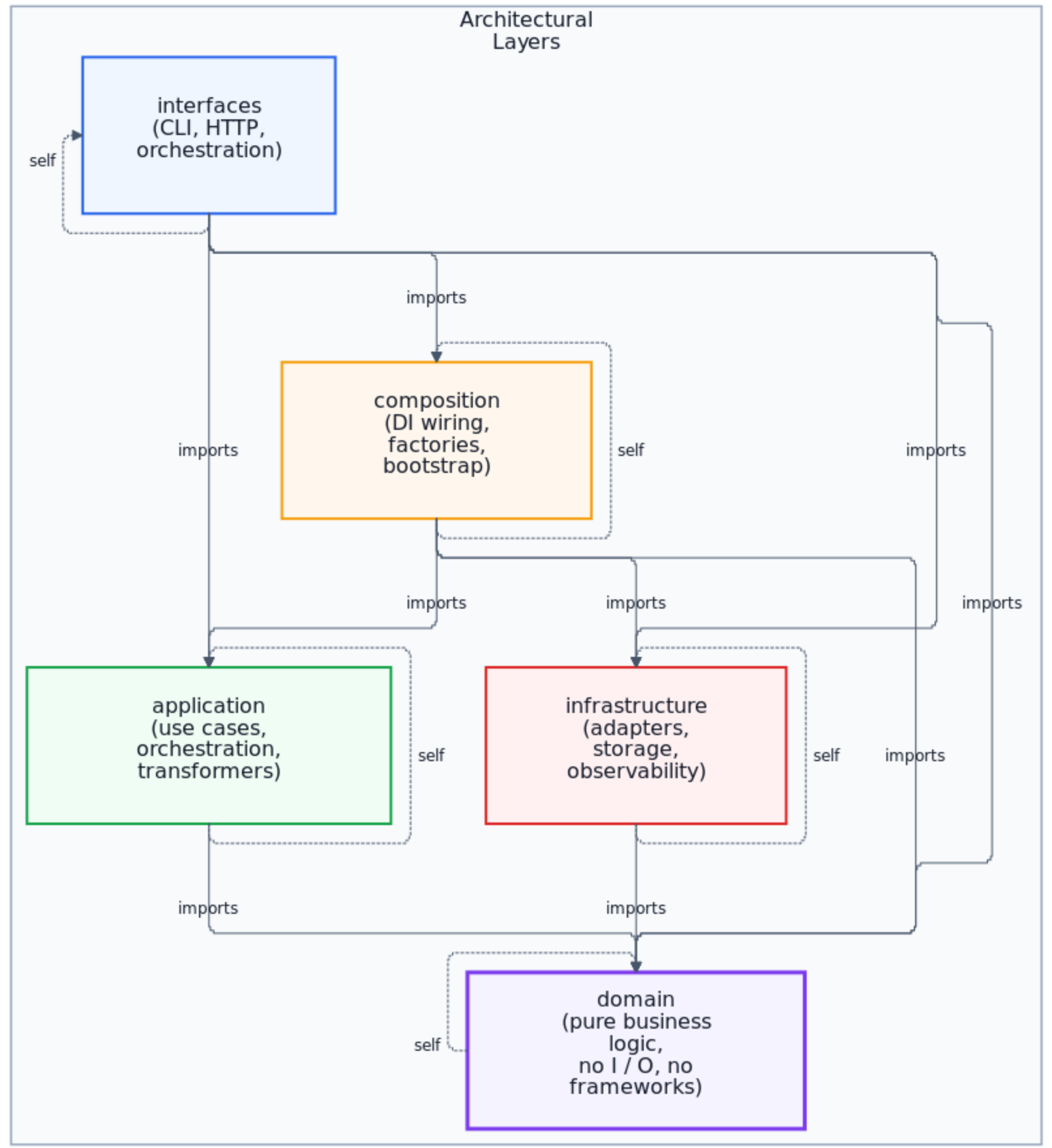


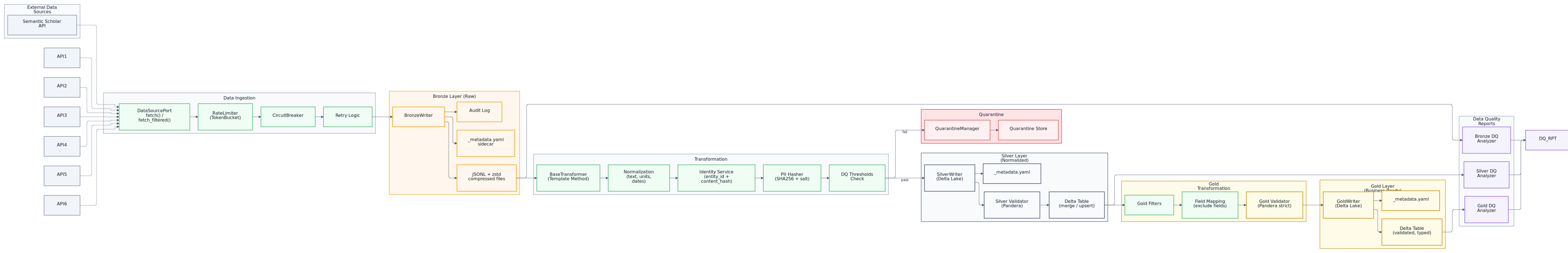


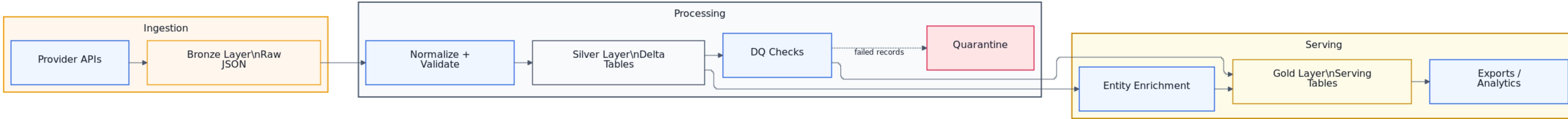


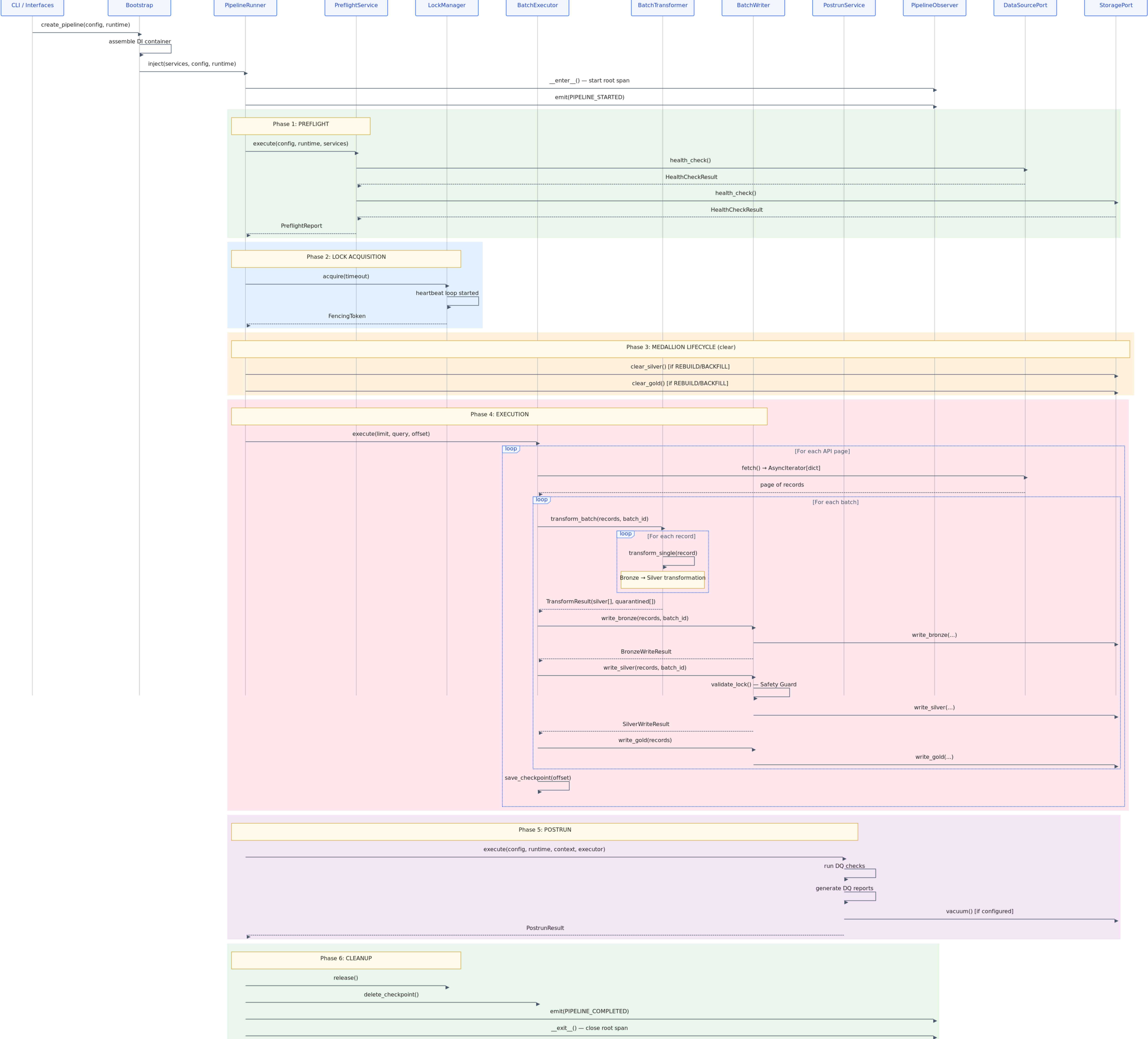


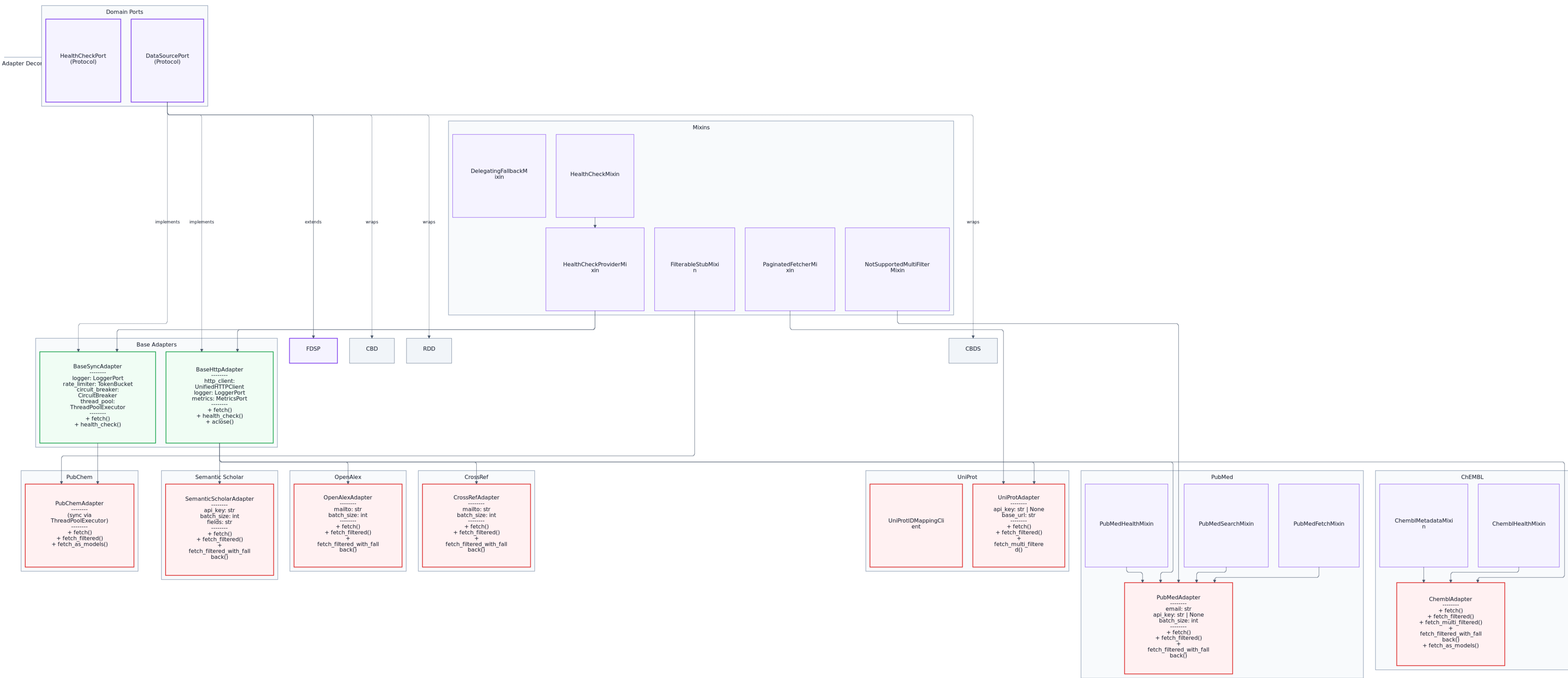


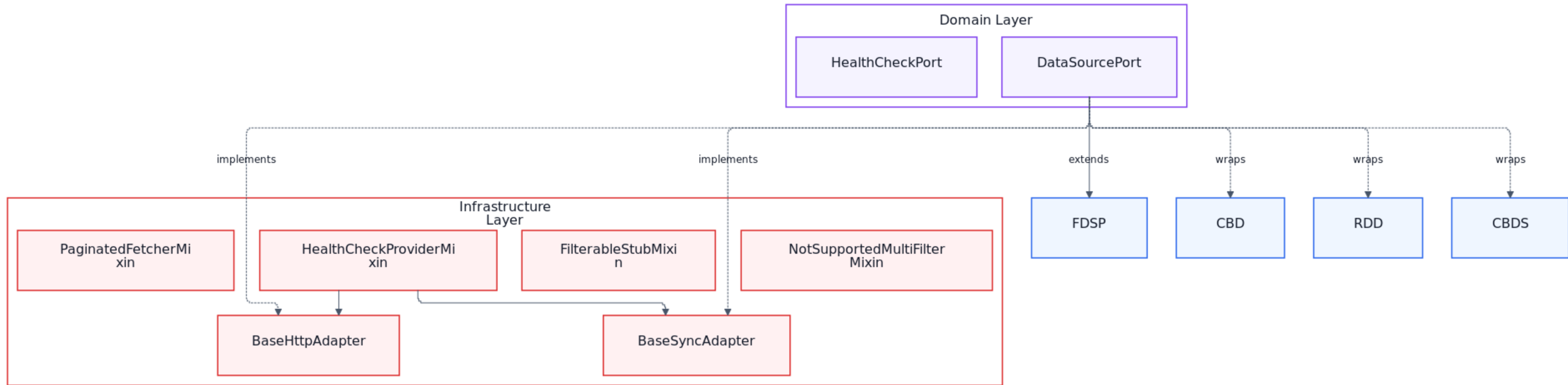


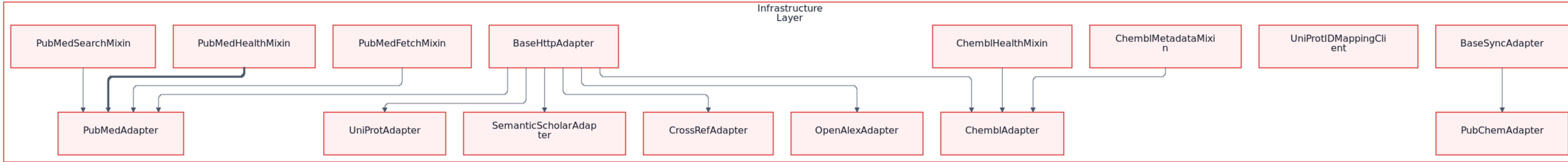


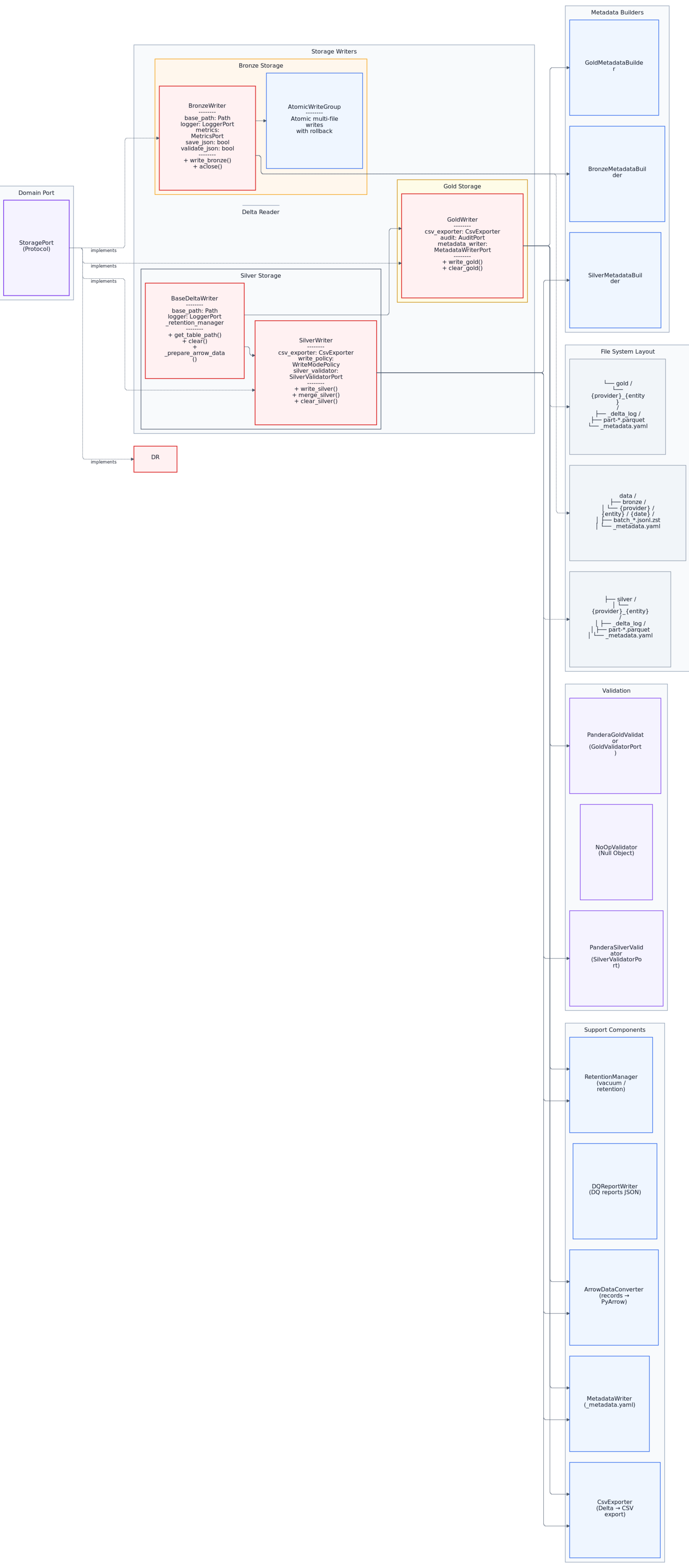


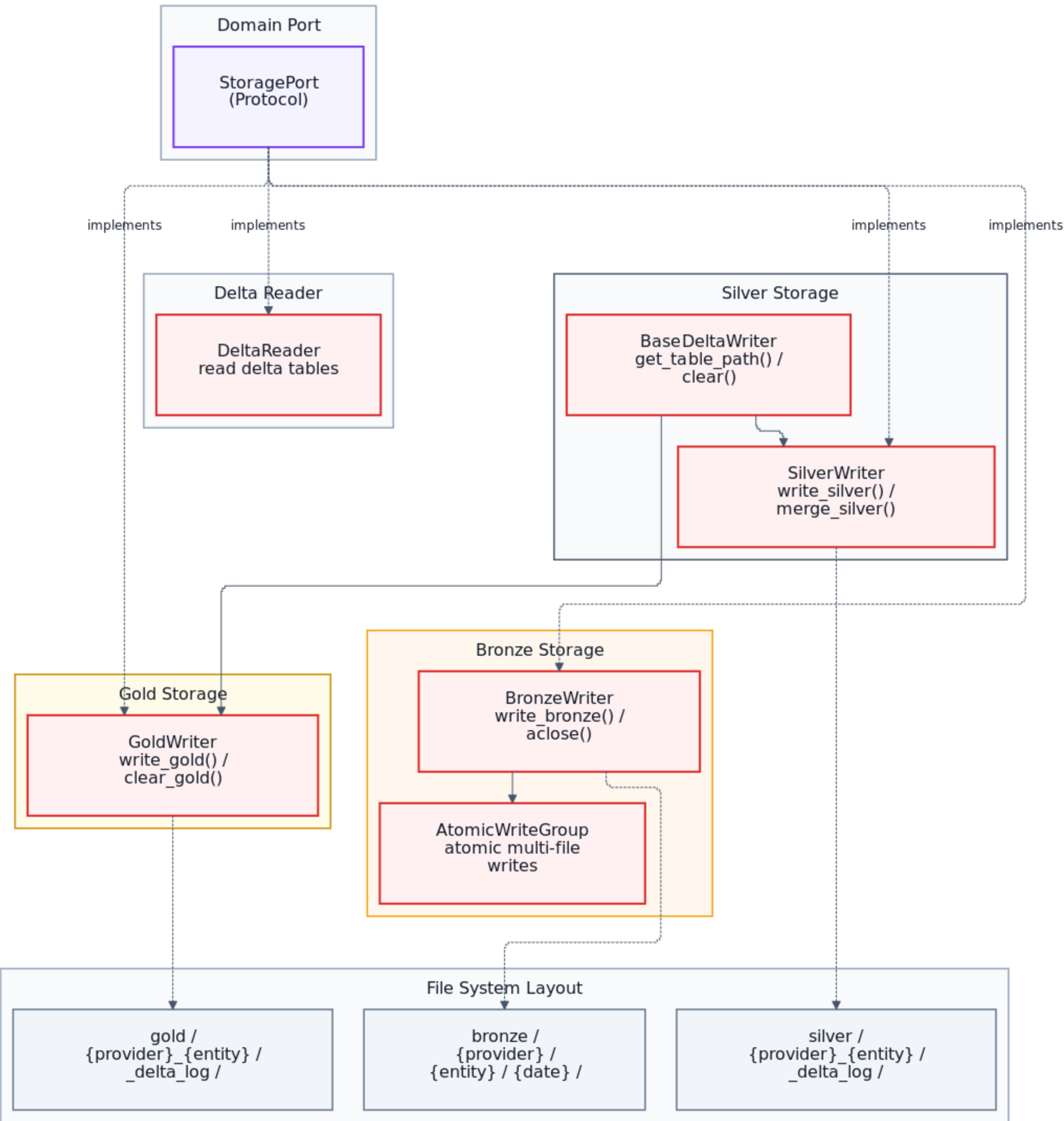


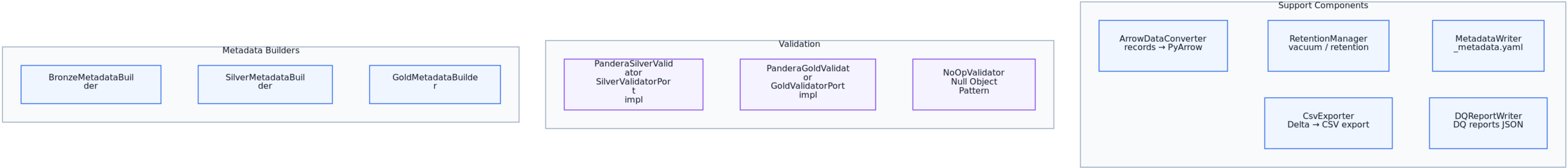


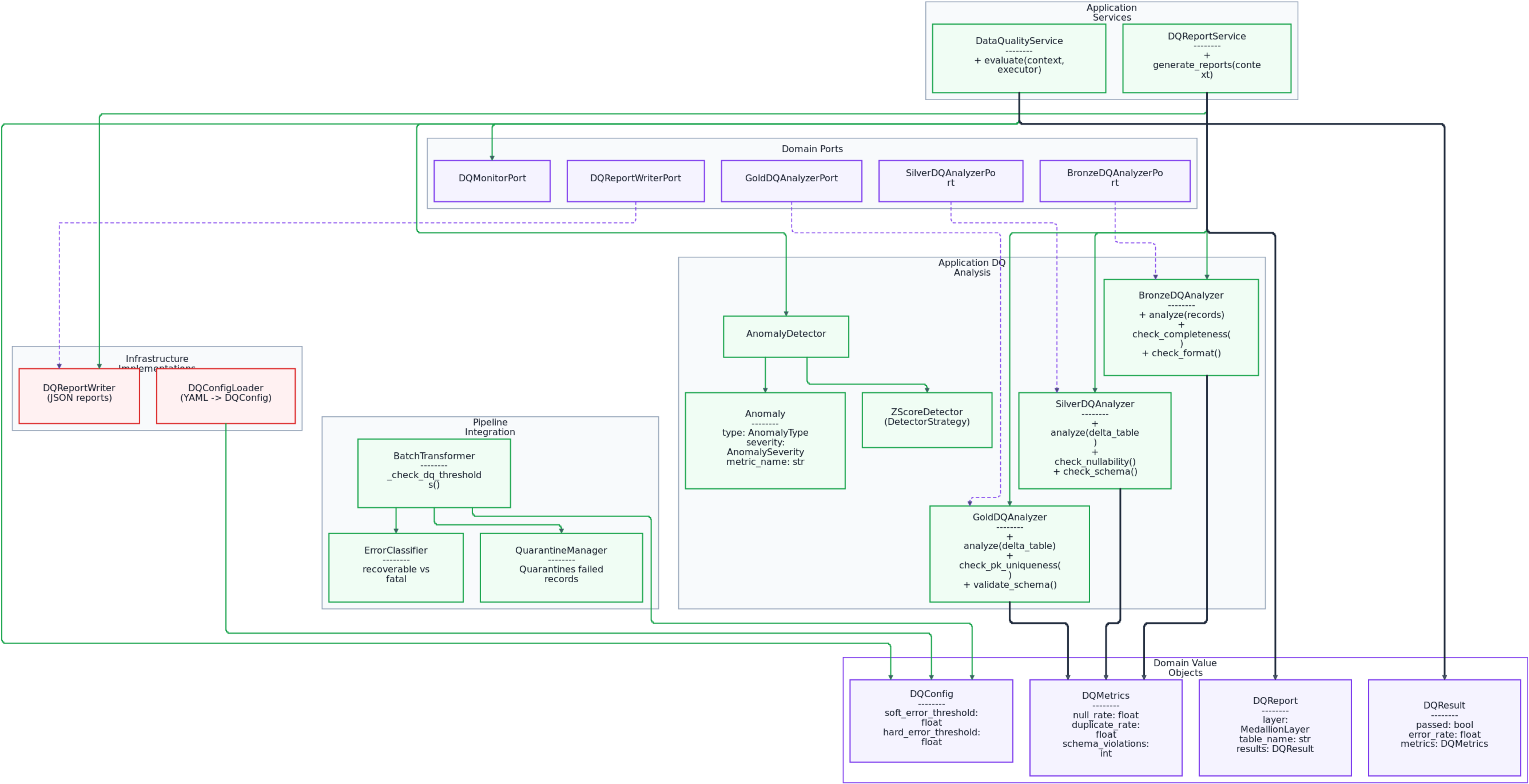


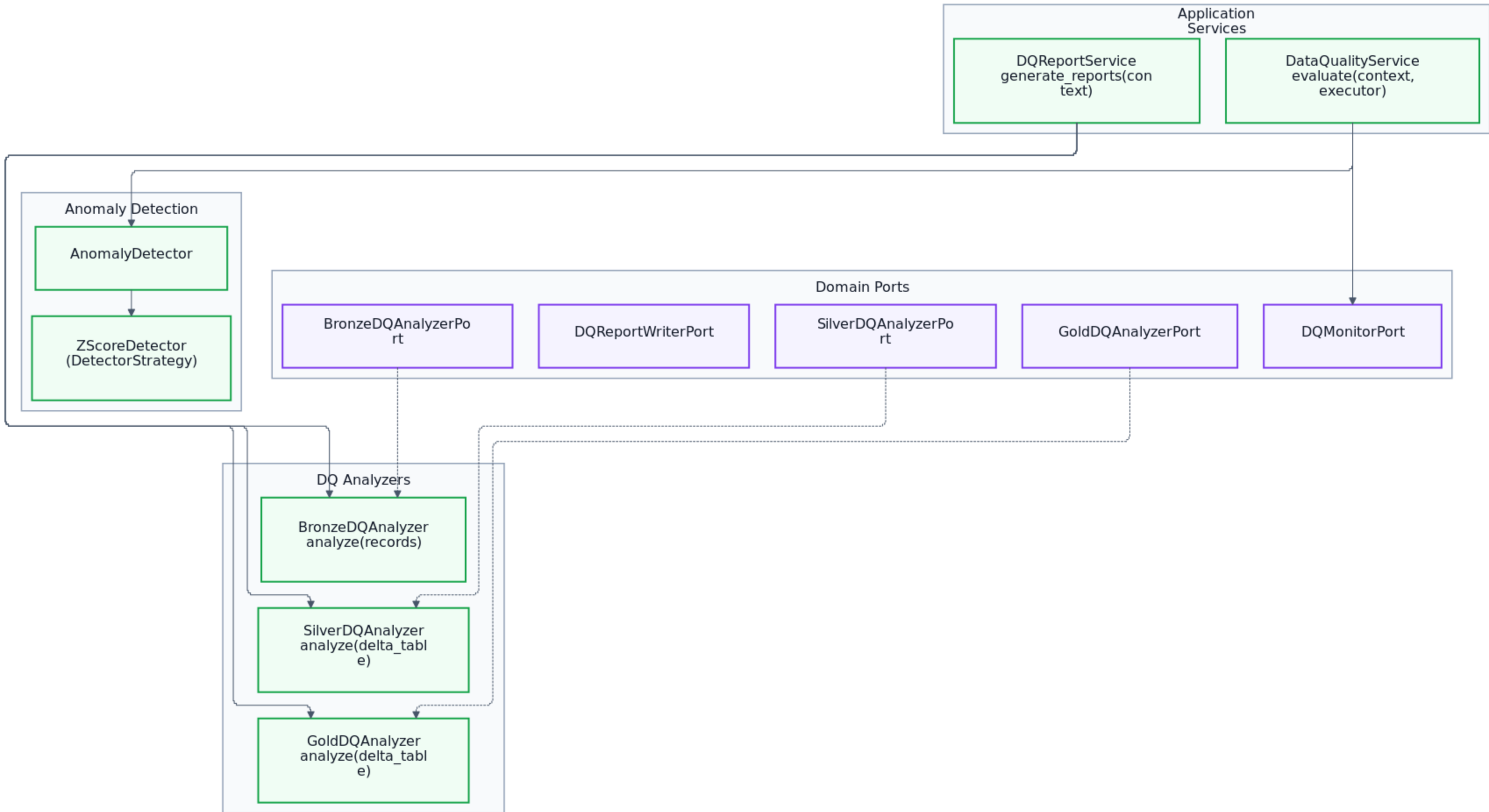


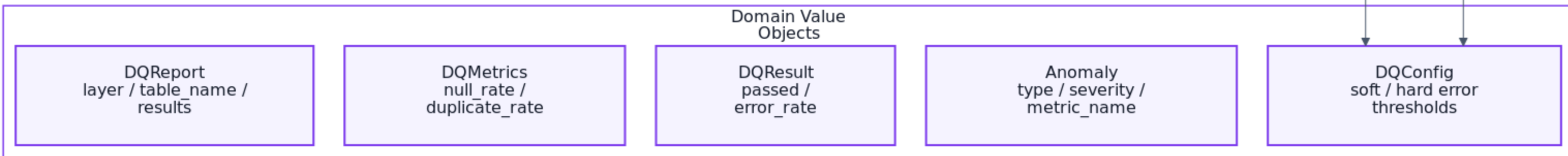
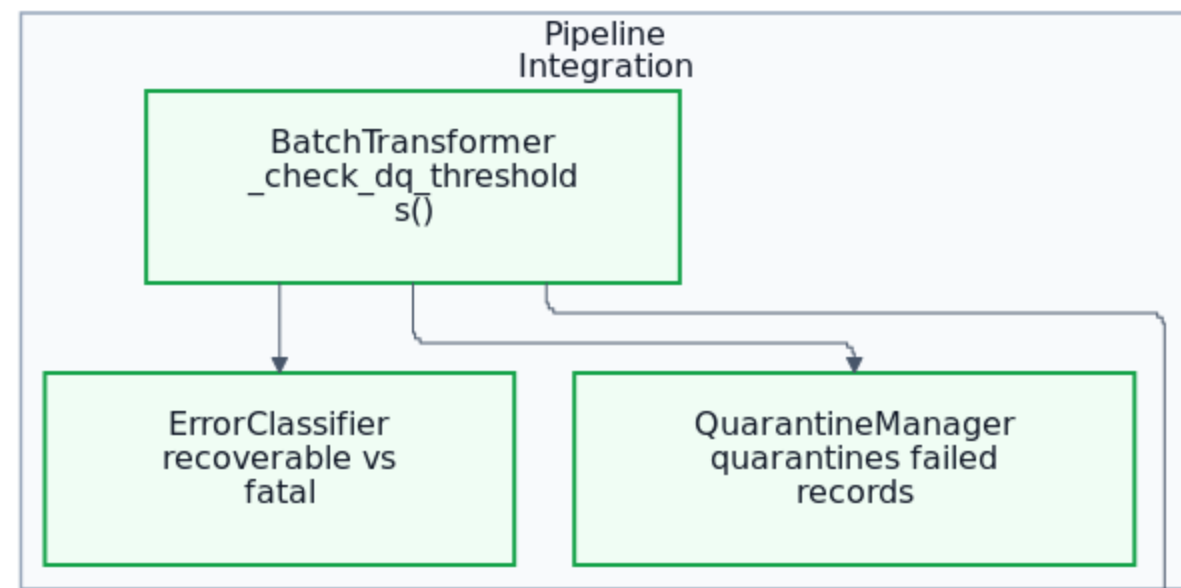
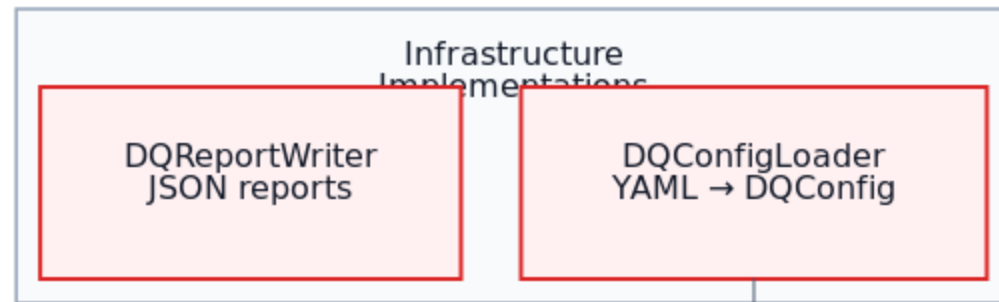


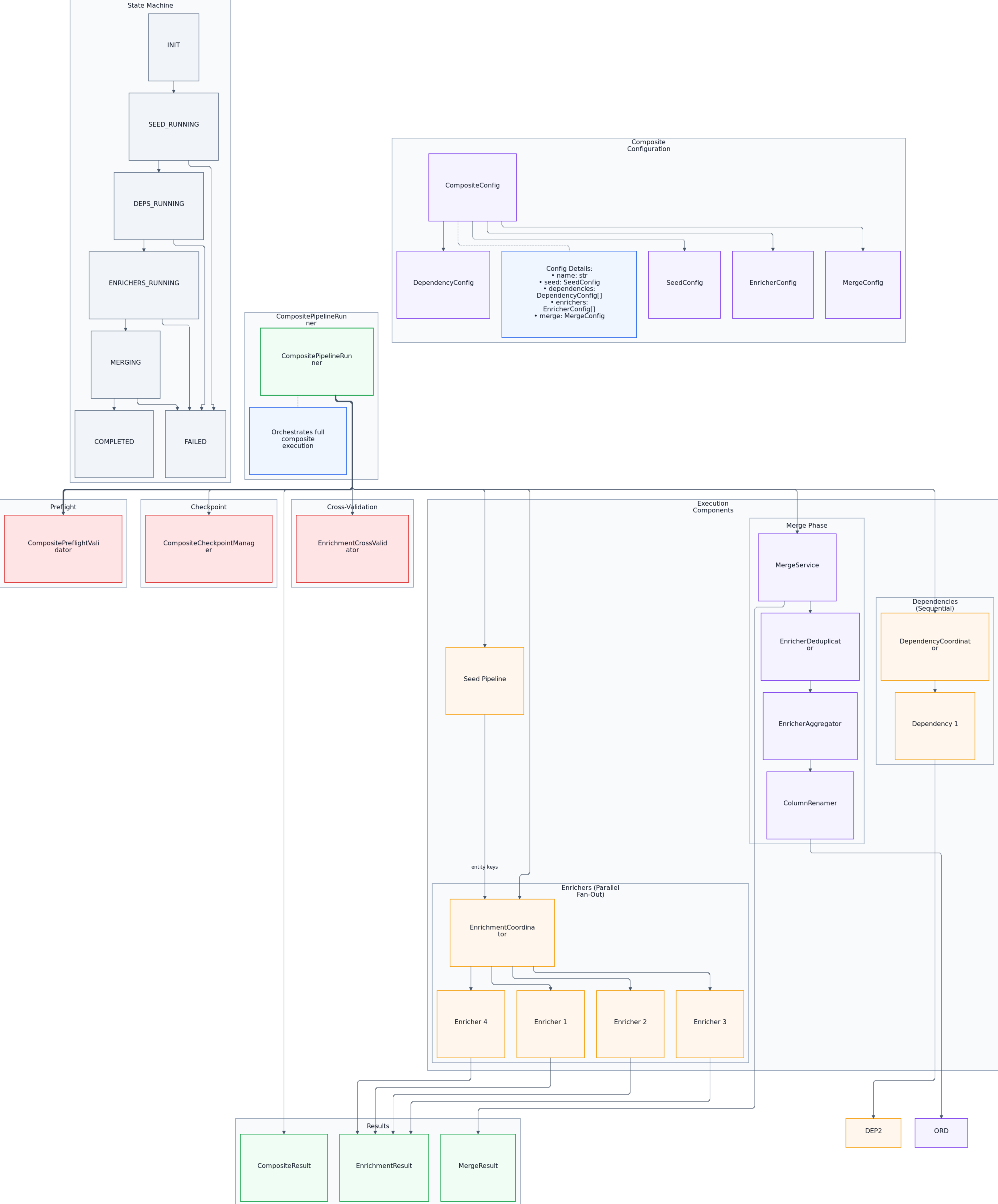


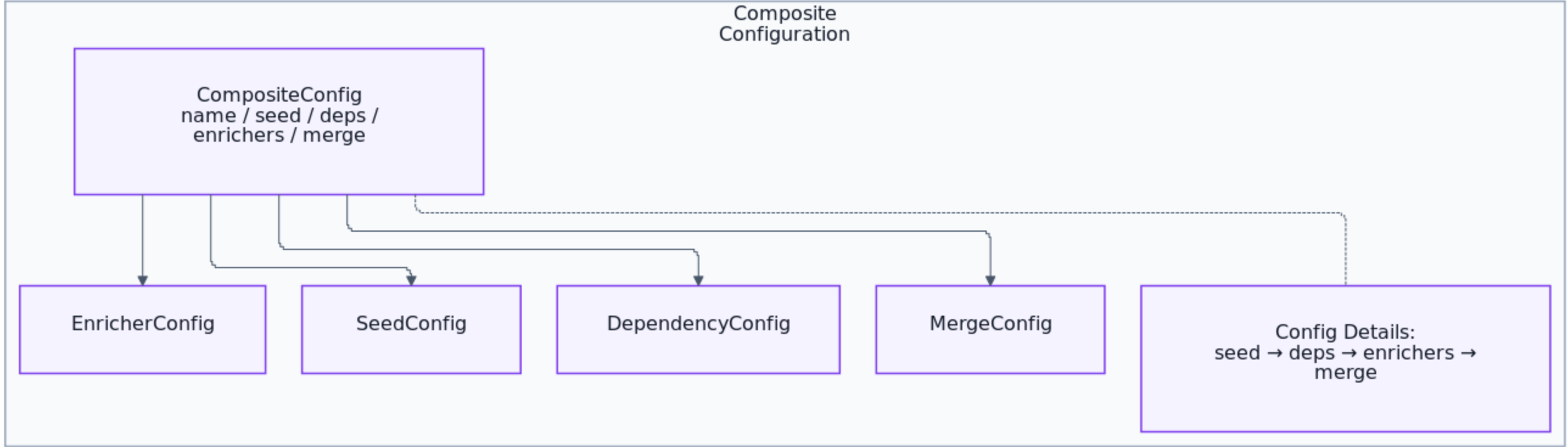
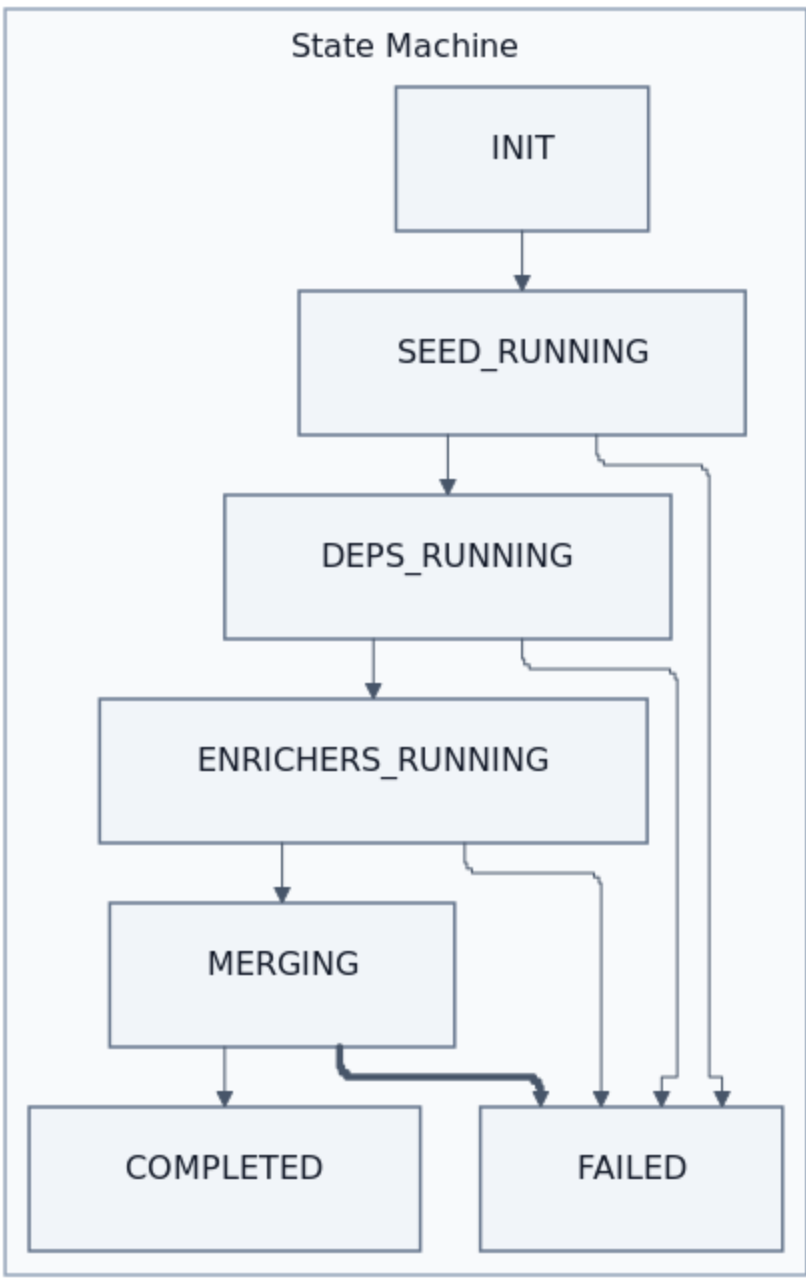












Composite Pipeline Runner

CompositePipelineRun
ner
orchestrates full
execution

Checkpoint

CheckpointManager

Cross-Validation

EnrichmentCrossValid
ator

Preflight

PreflightValidator

Execution
Components

Seed Pipeline

entity keys

Enrichers (Parallel
Fan-Out)

EnrichmentCoordina
tor

Enricher 1

Enricher 2

Enricher 3

Enricher 4

Dependencies
(Sequential)

DependencyCoordinat
or

Dependency 1

Merge Phase

MergeService

EnricherDeduplicat
or

EnricherAggregator

ColumnRenamer

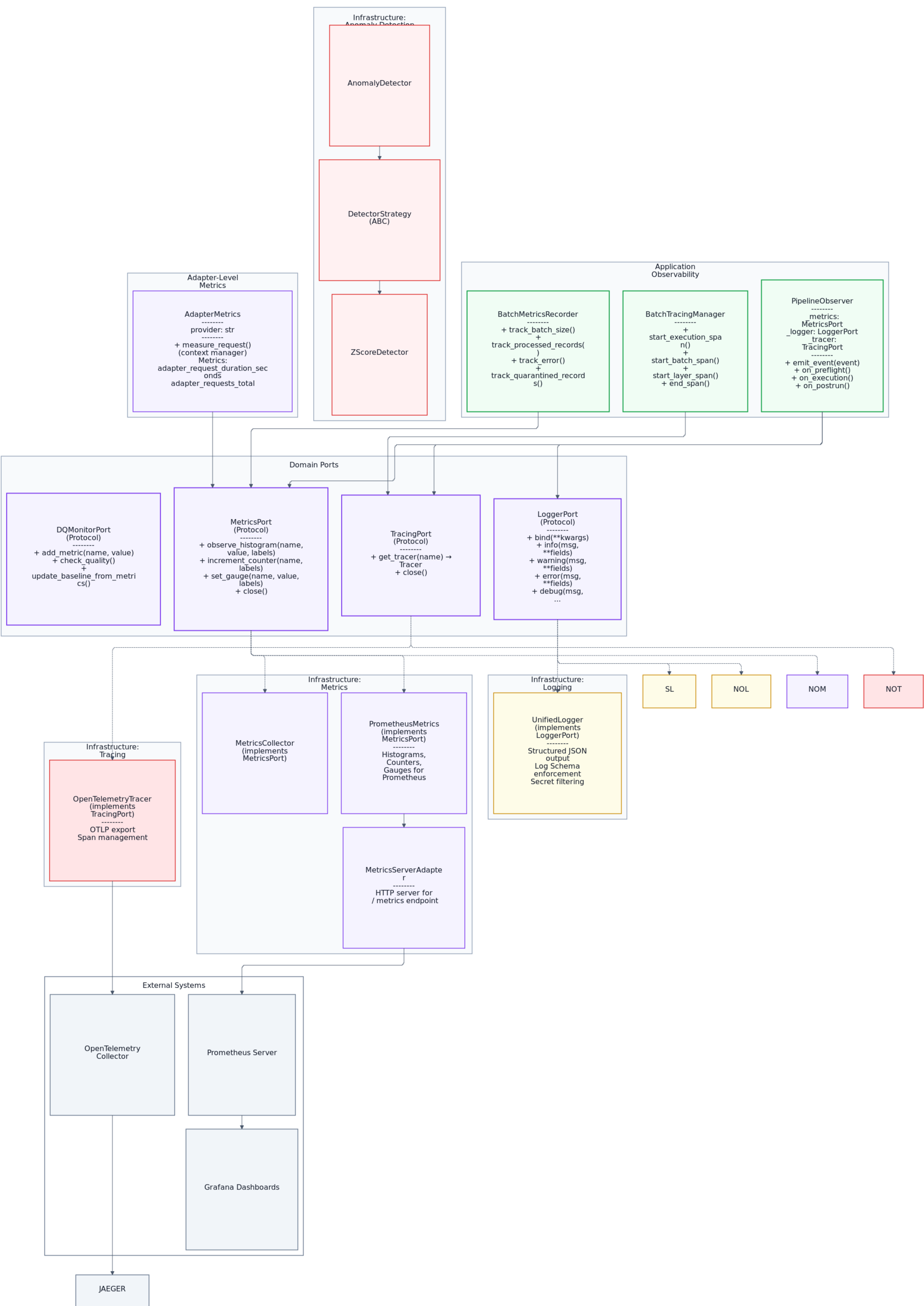
ColumnOrderer

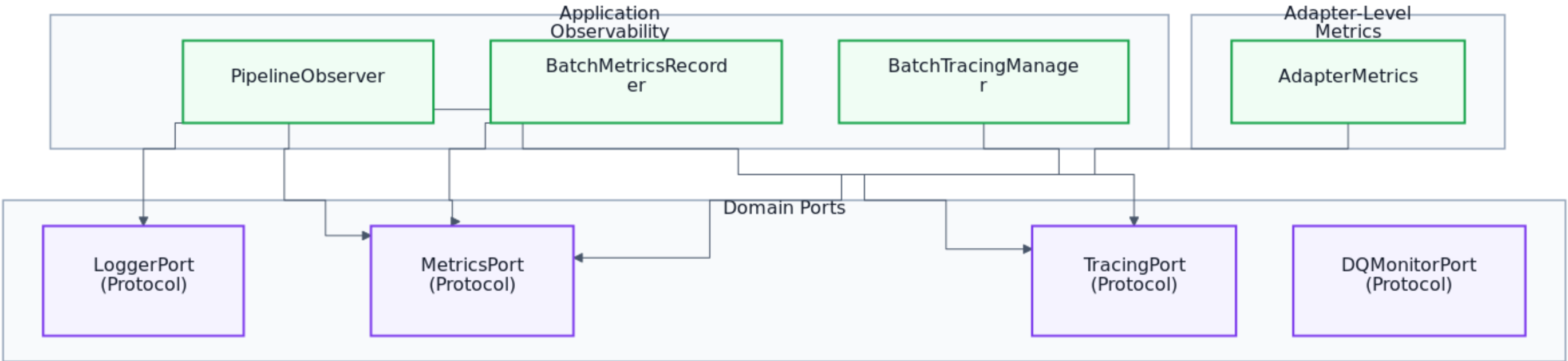
Results

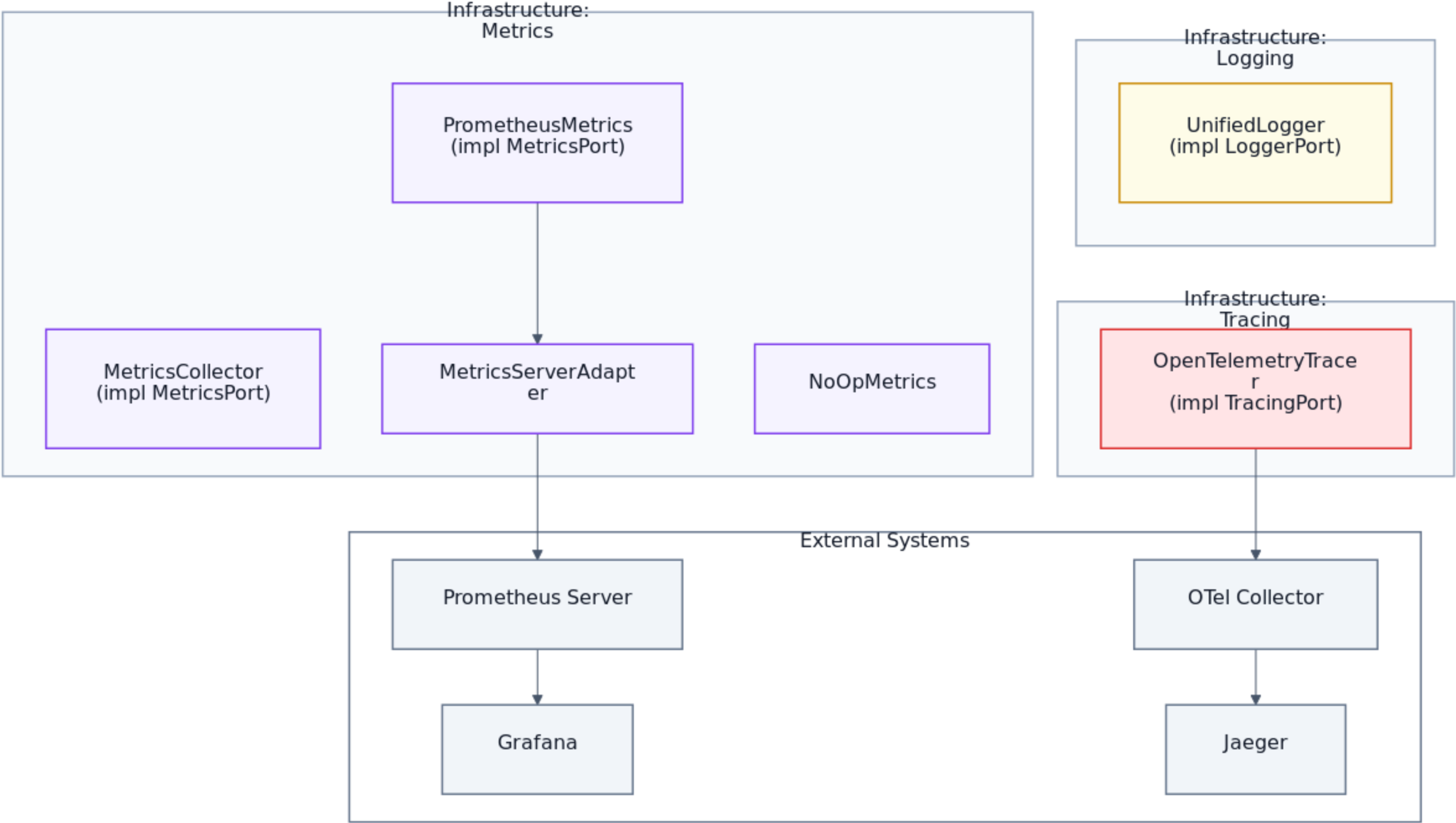
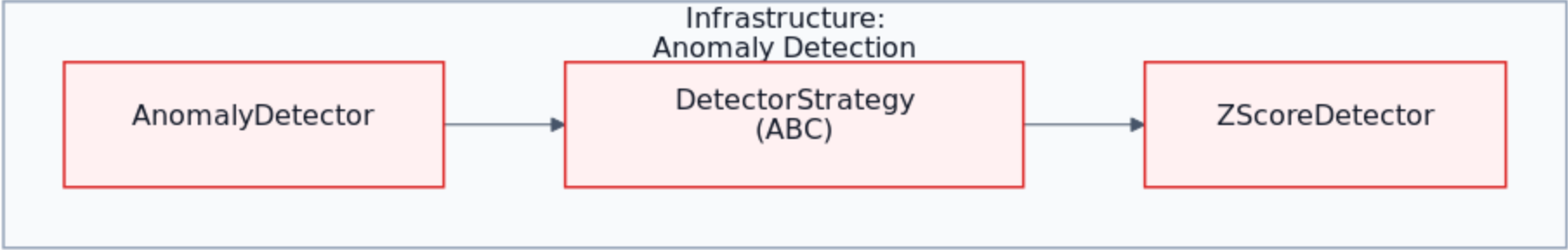
EnrichmentResult

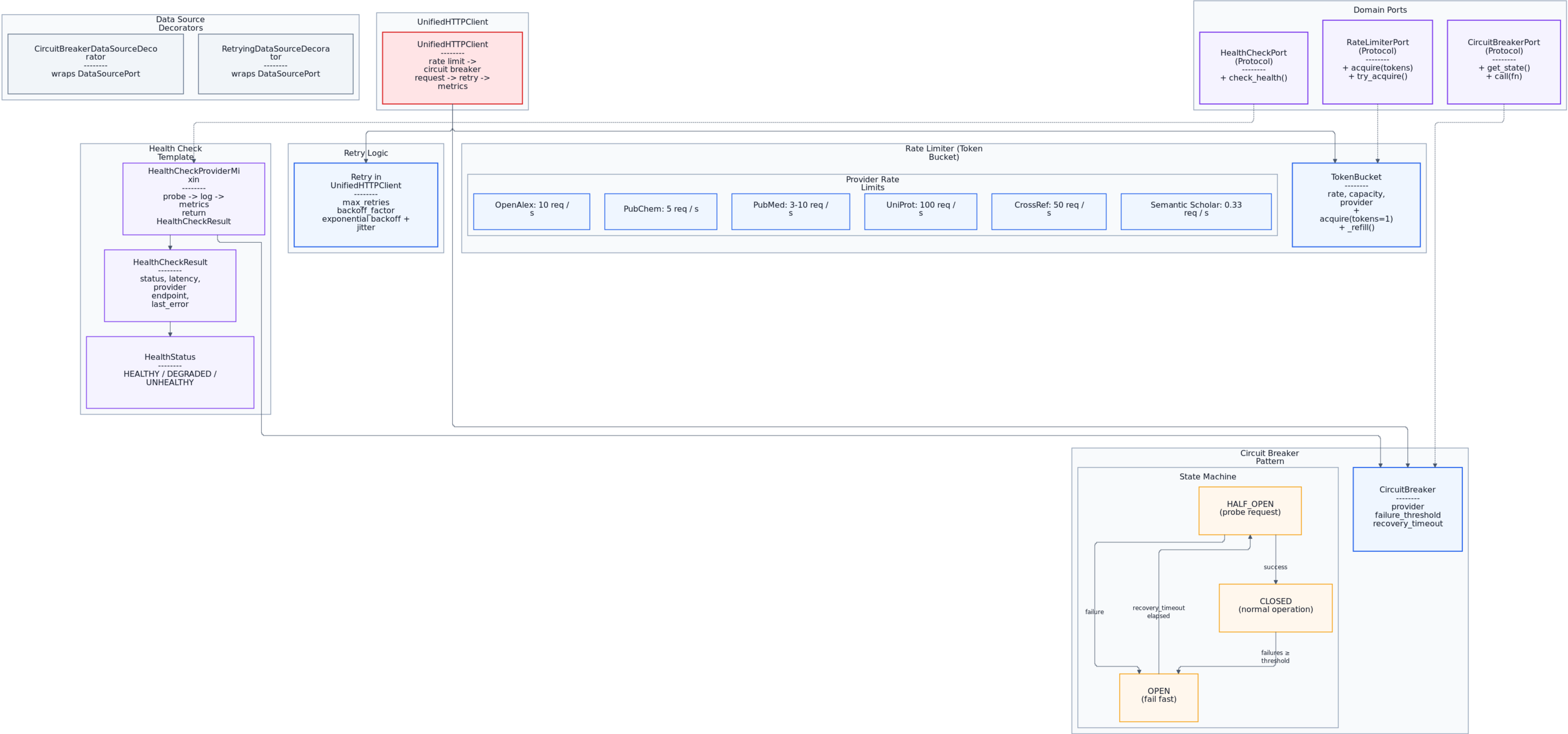
MergeResult

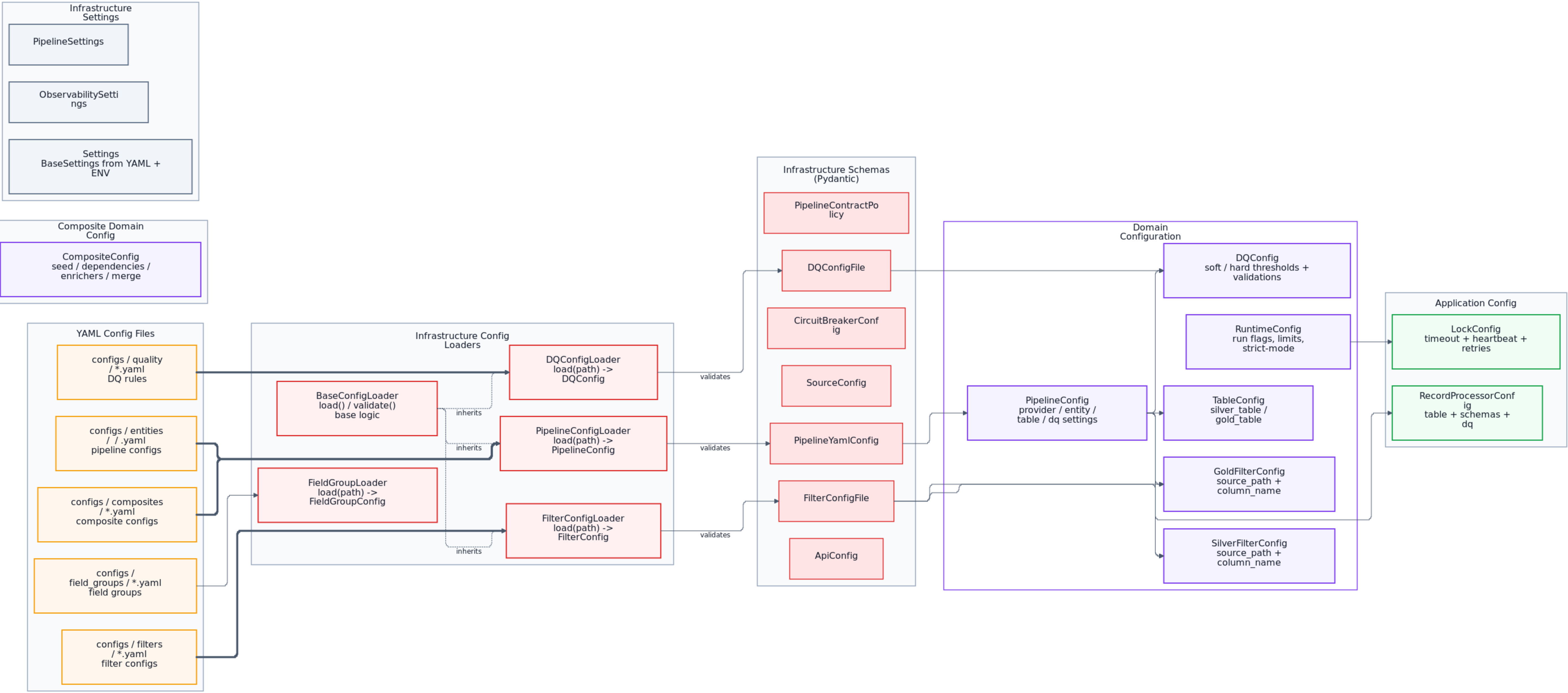
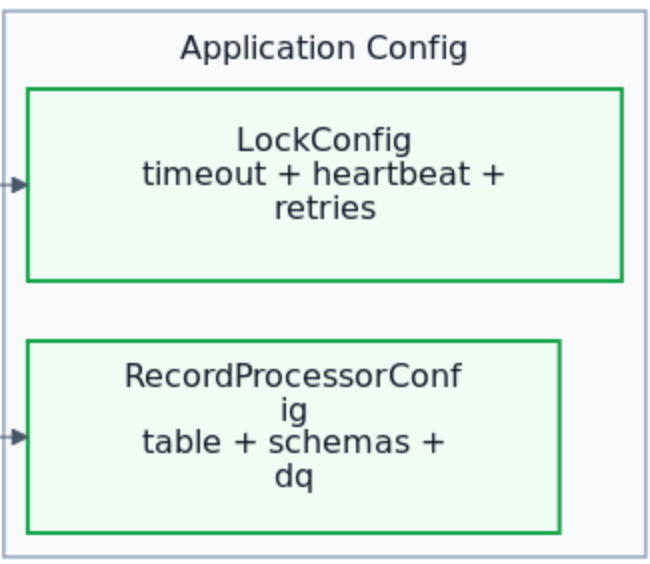
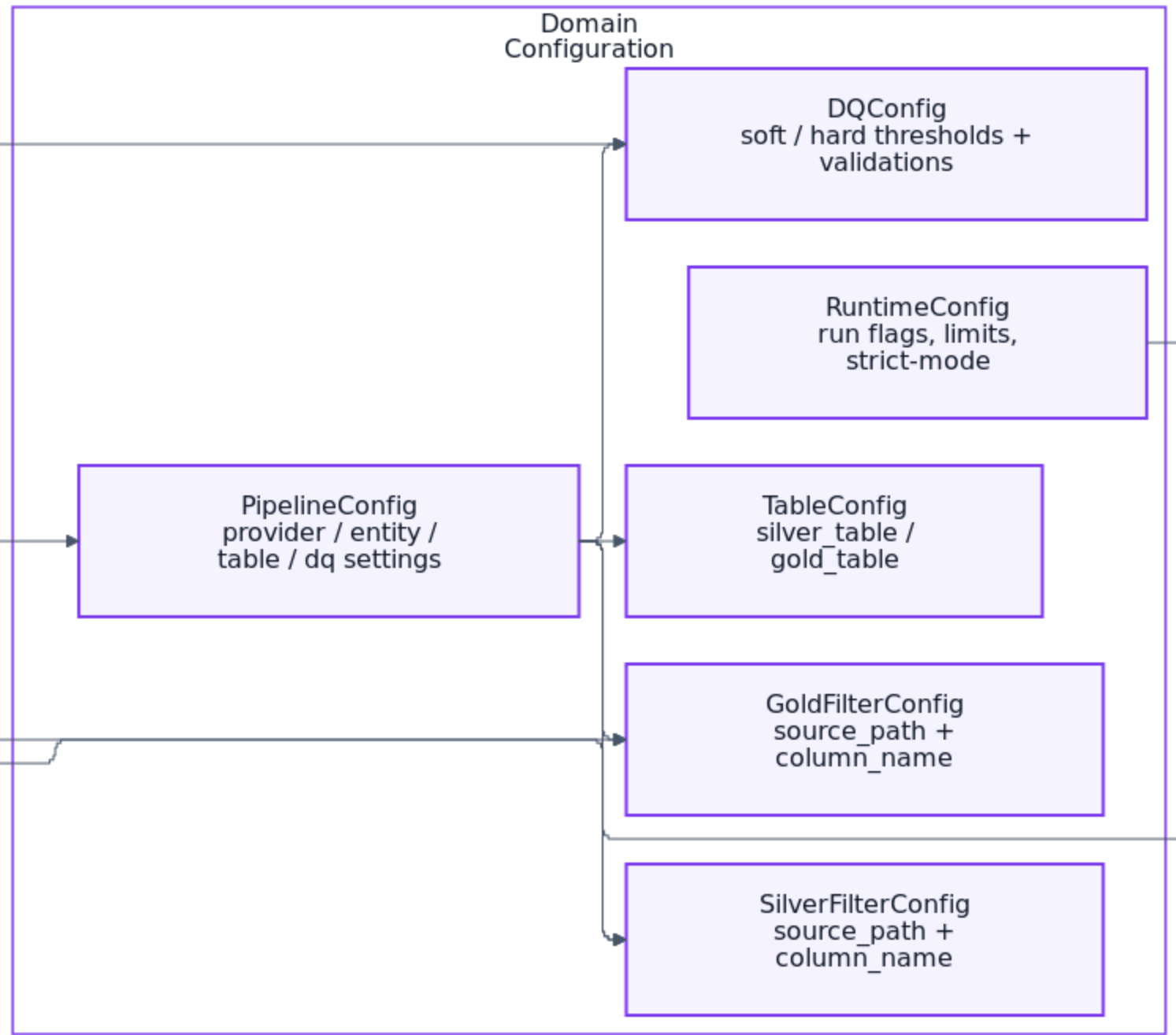
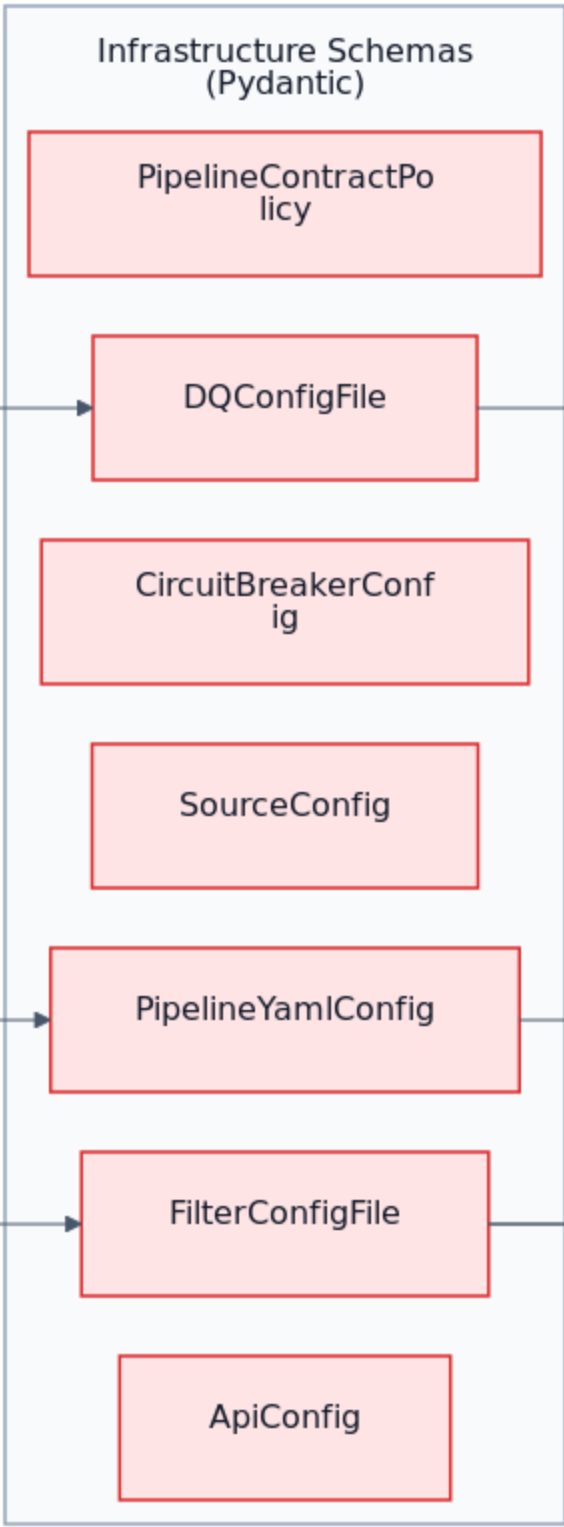
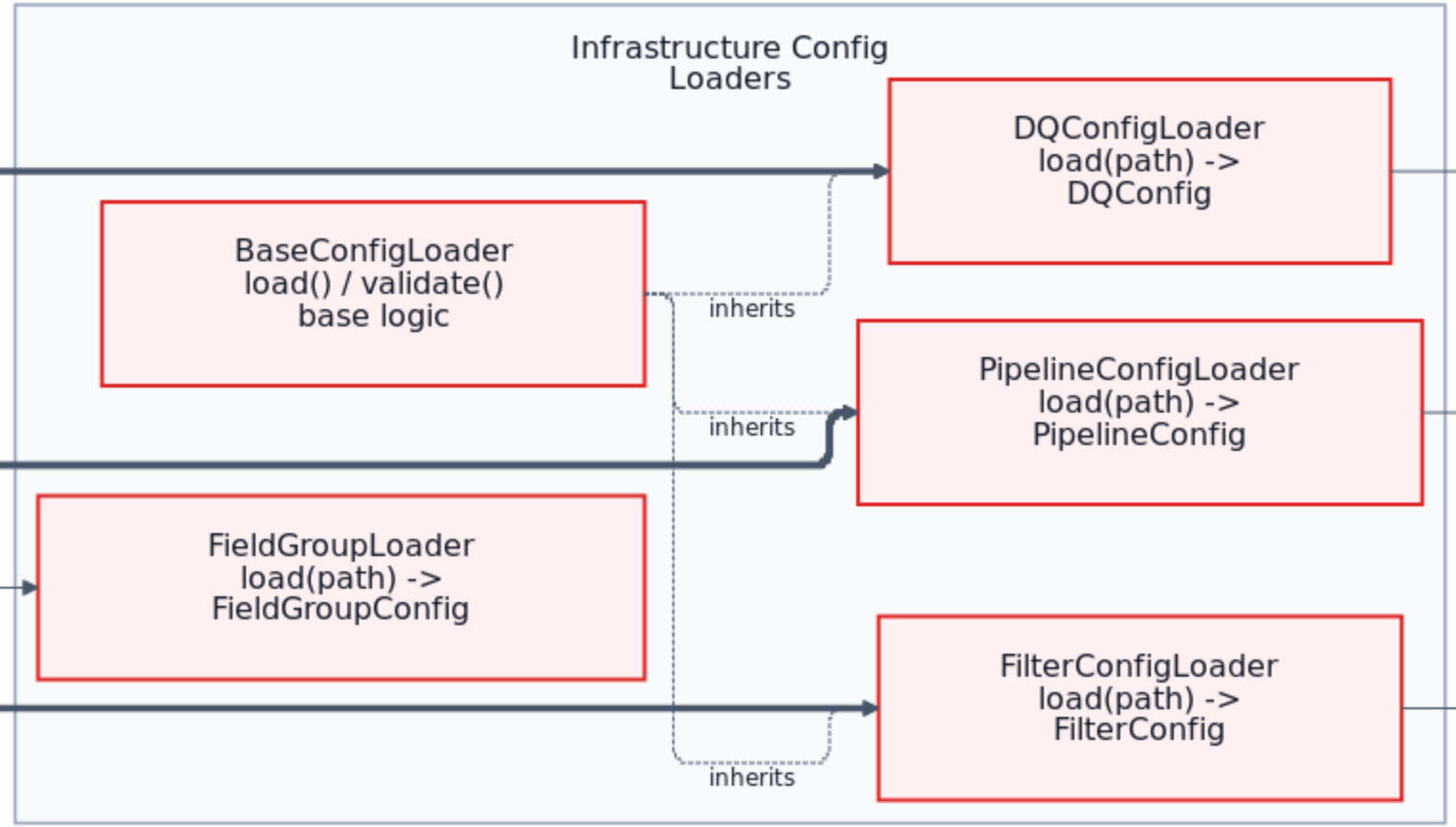
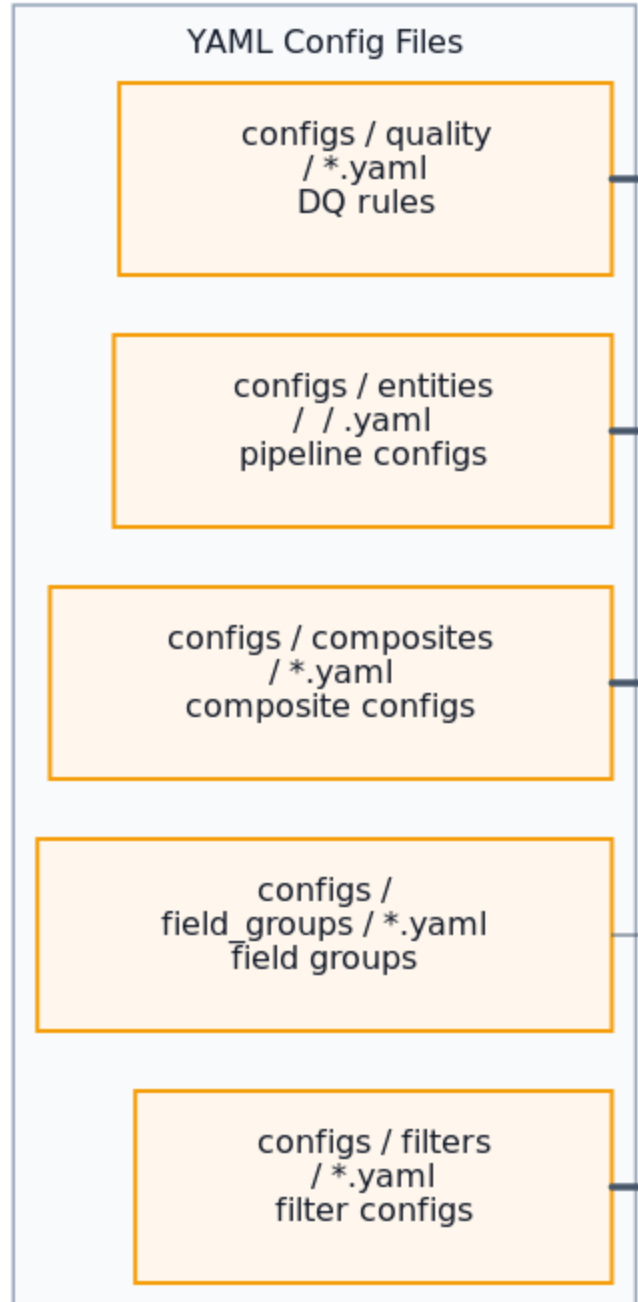
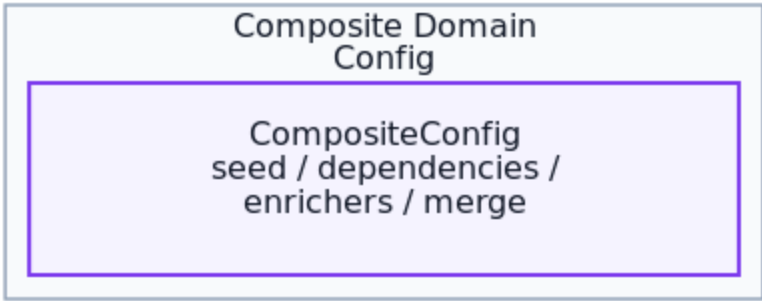
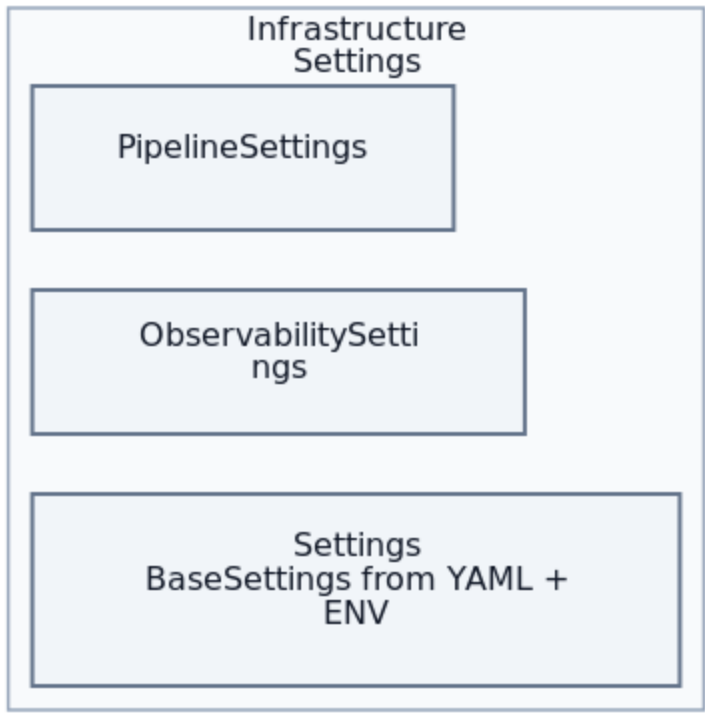
CompositeResult

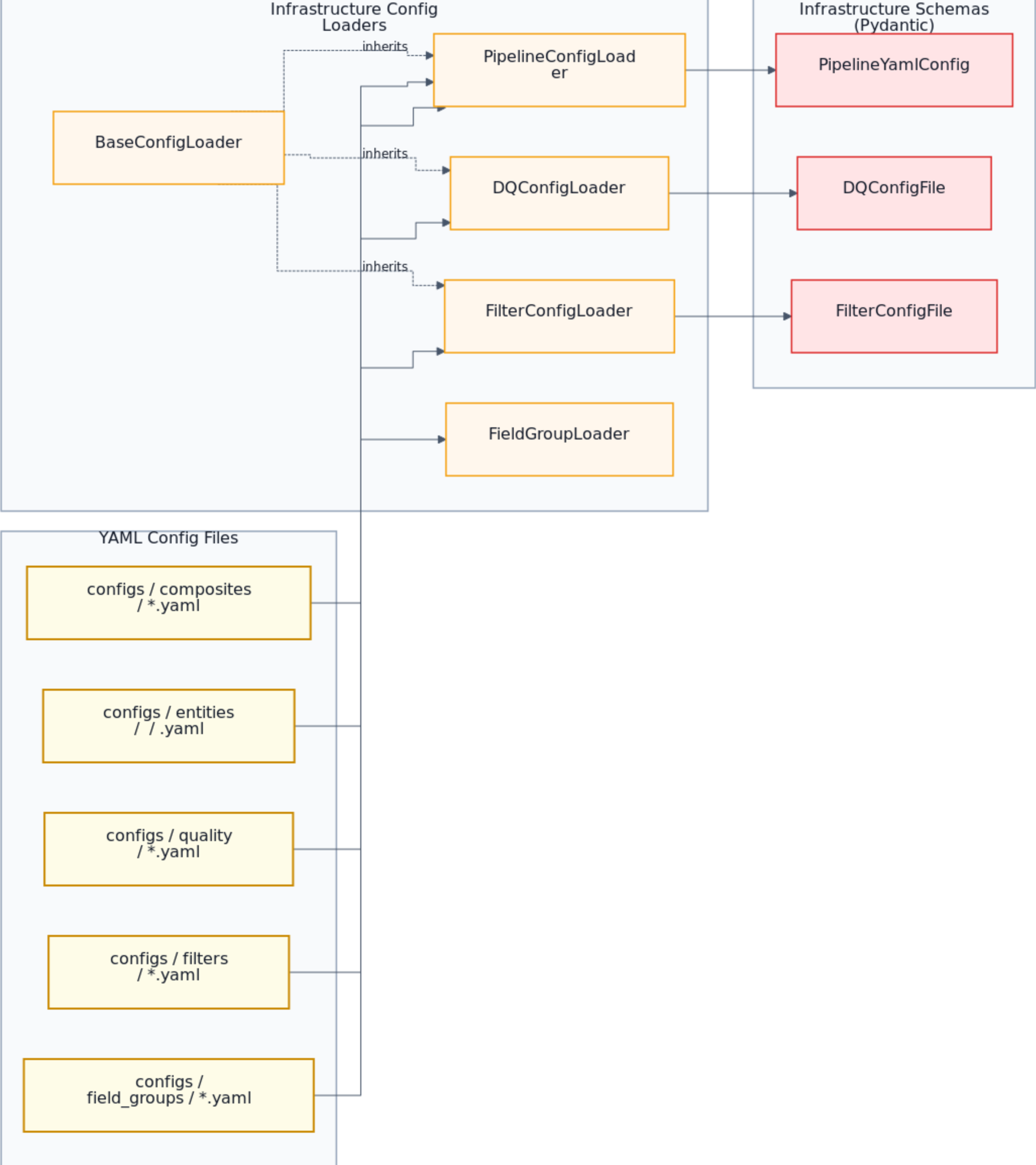


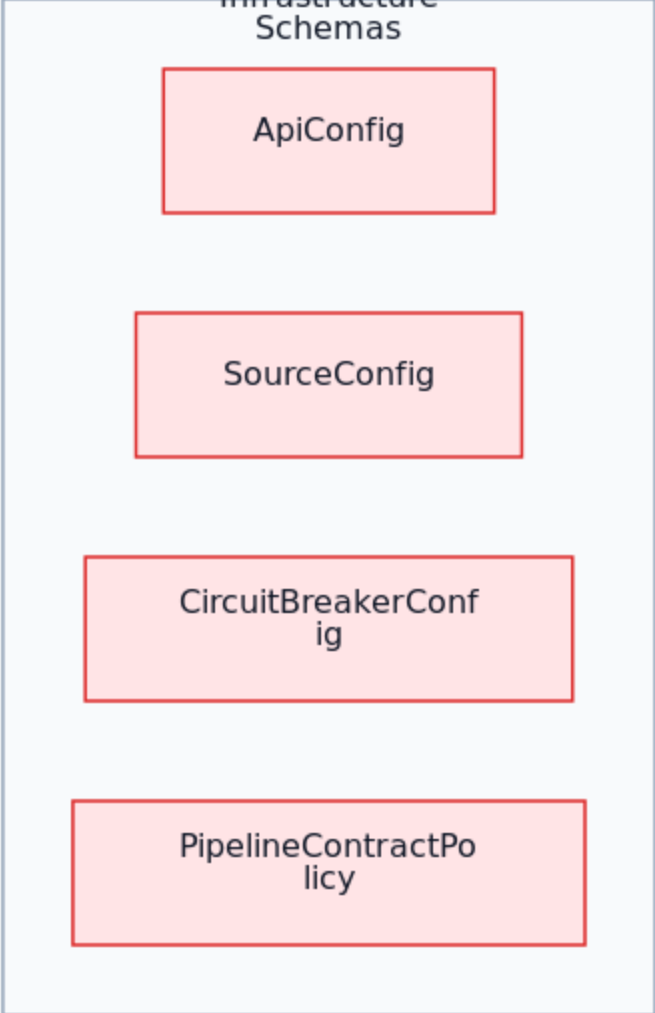
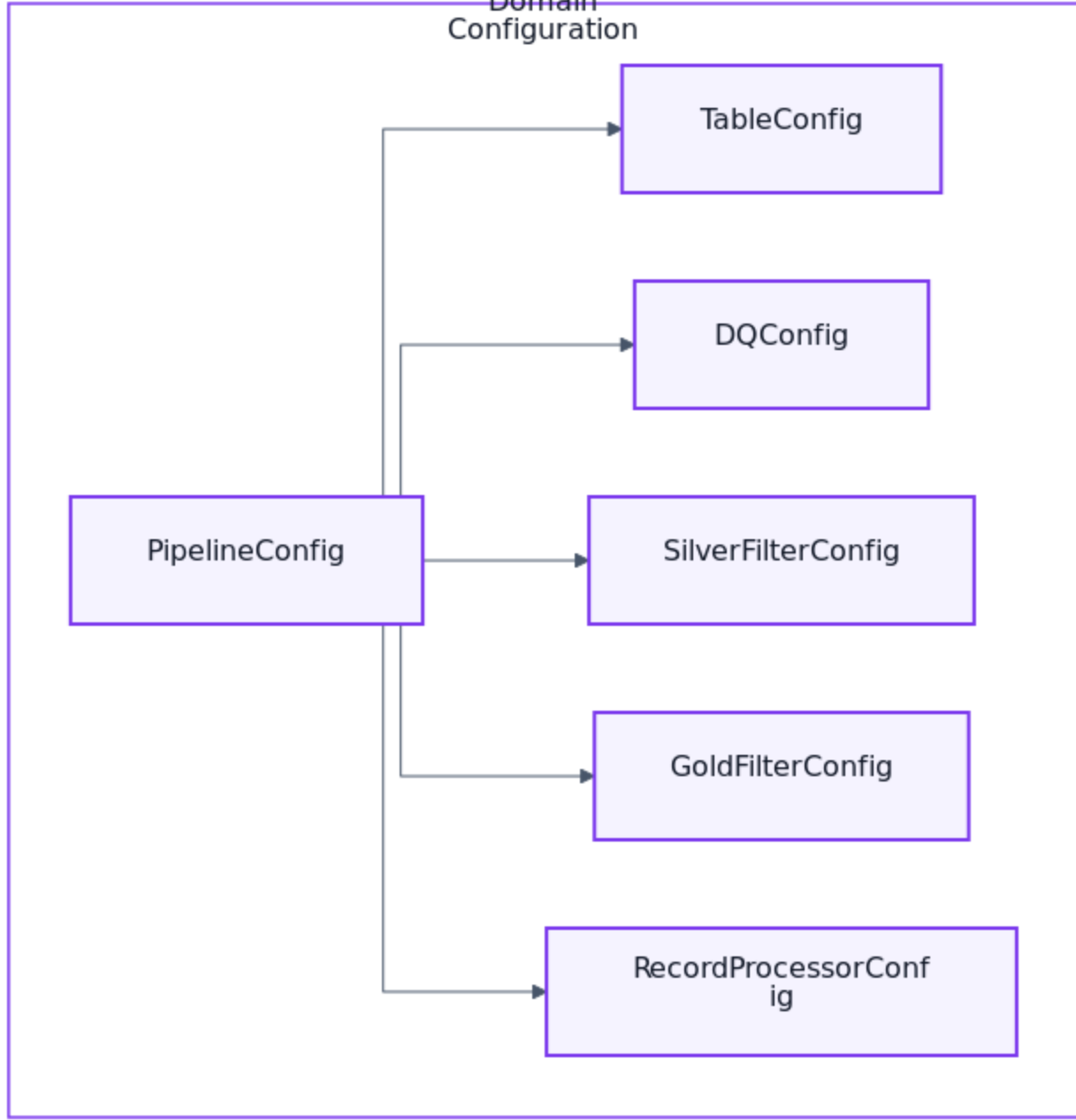
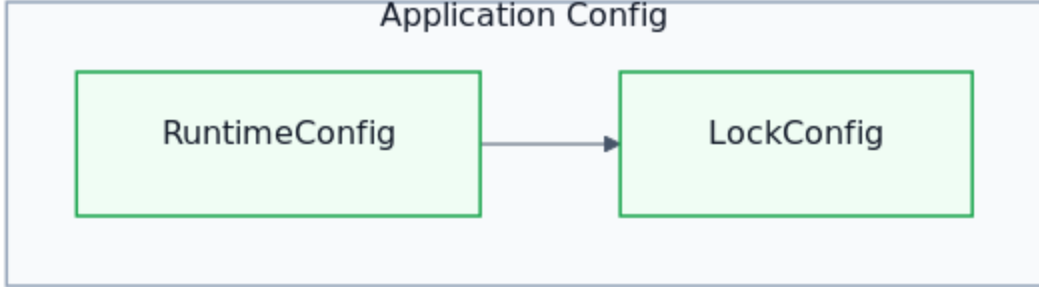
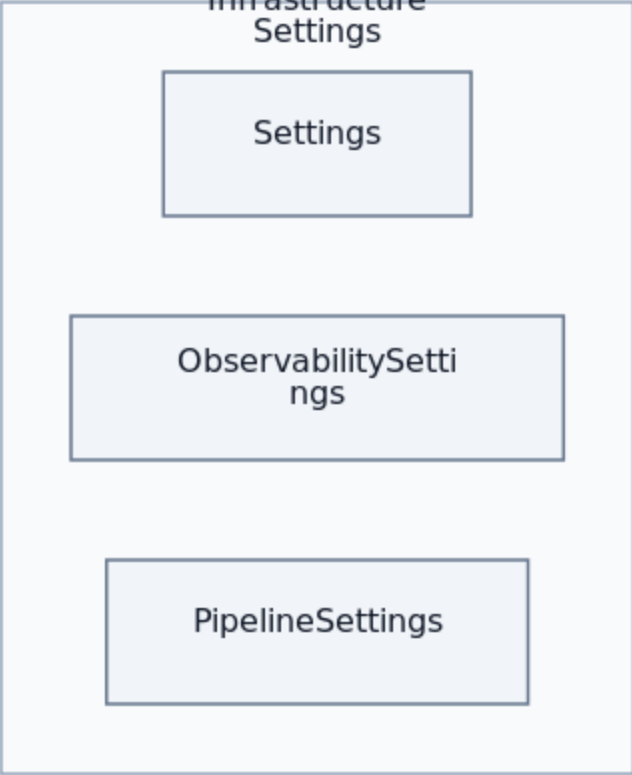


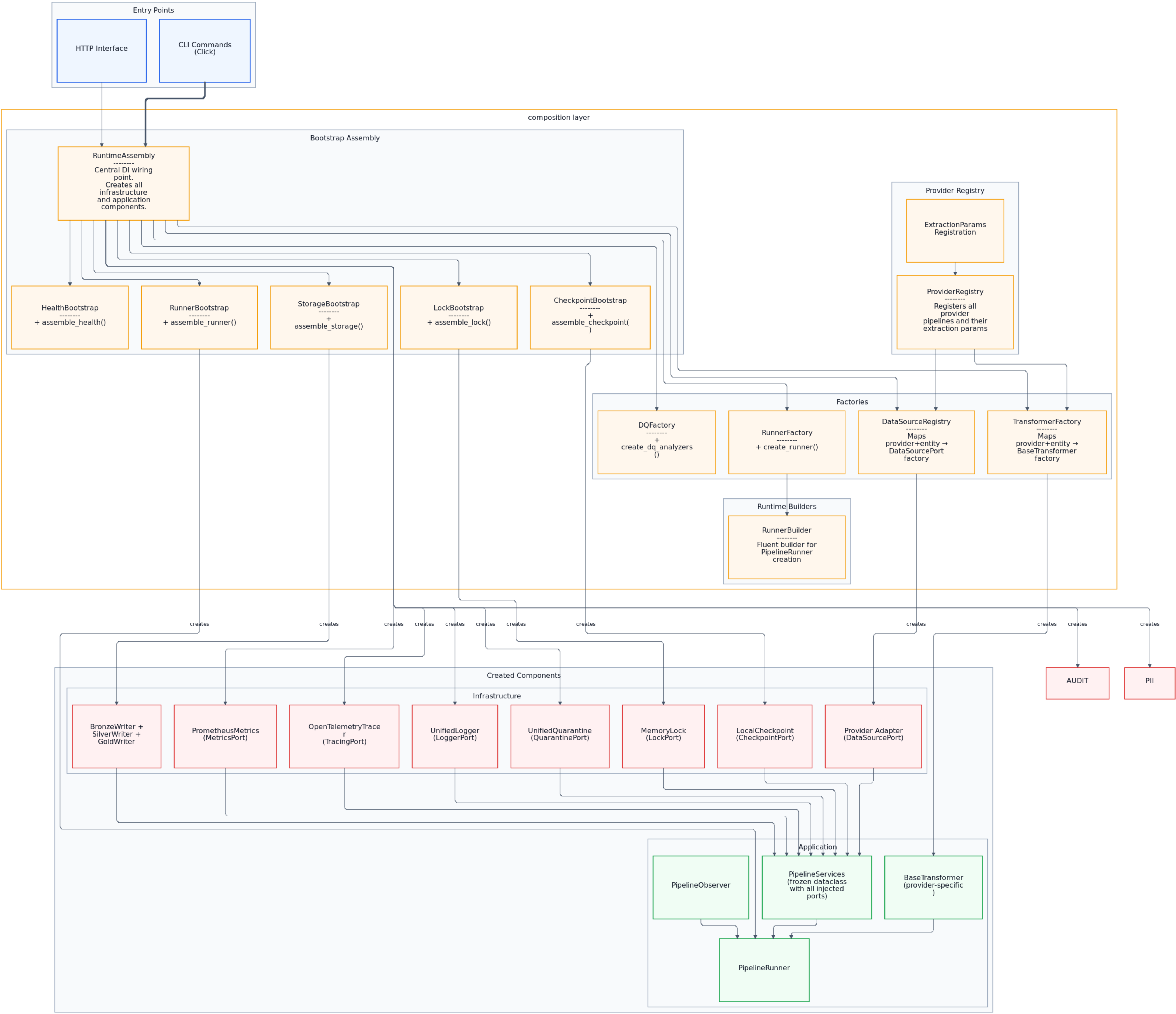


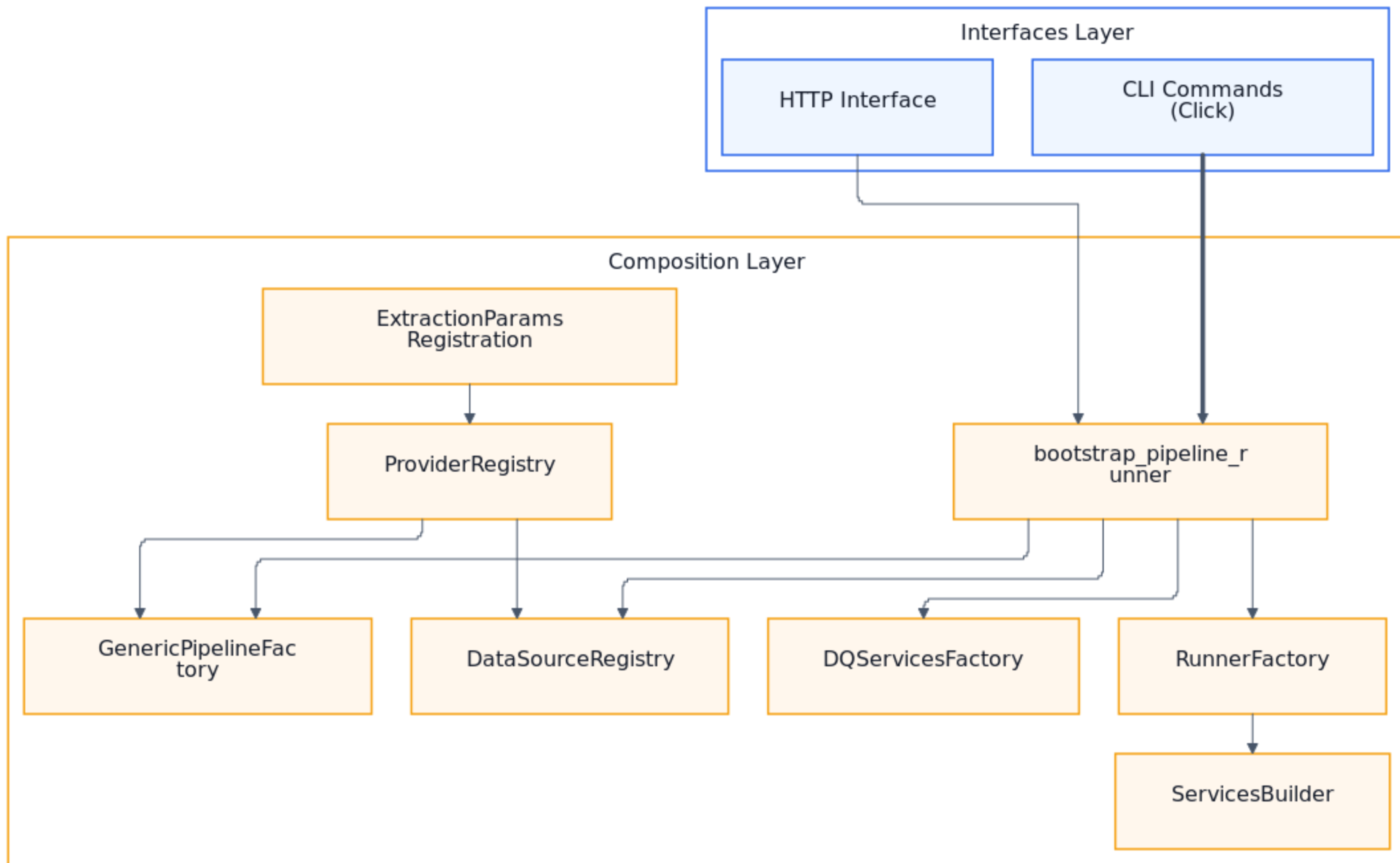


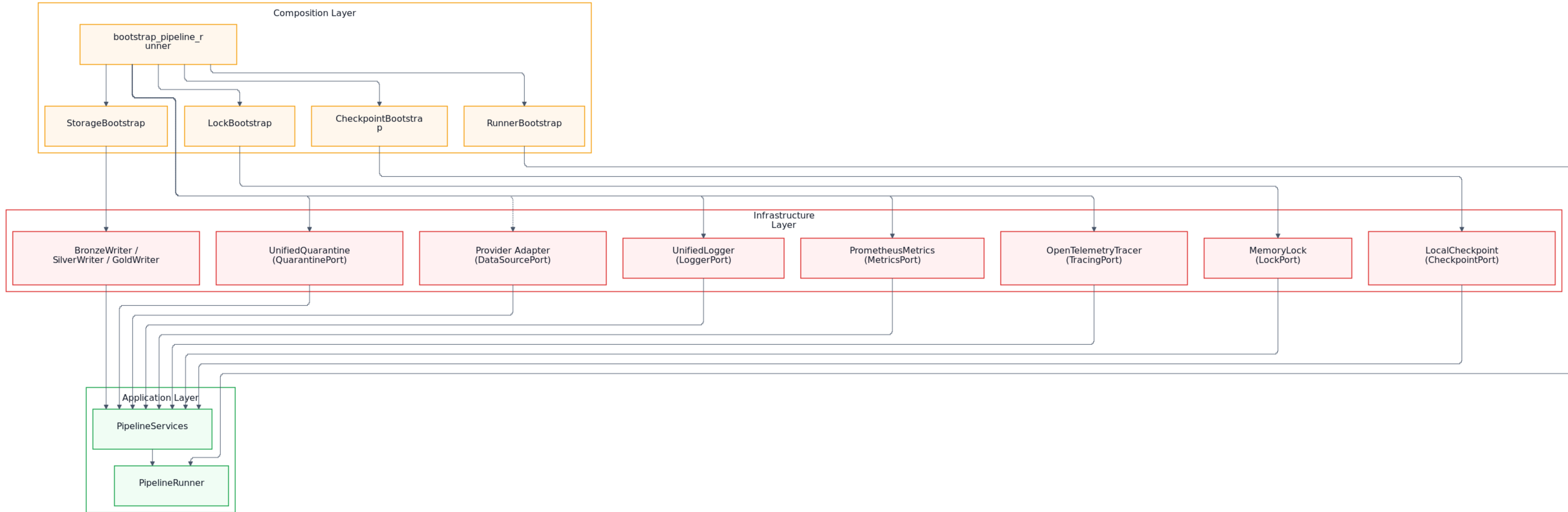


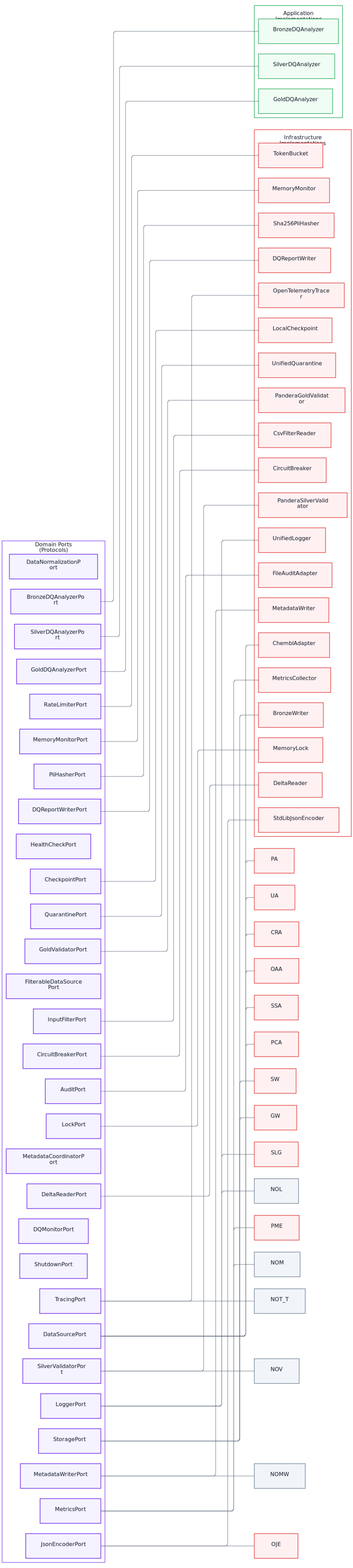












Domain Ports

DeltaReaderPort

FilterableDataSourcePort

InputFilterPort

StoragePort

DataSourcePort

Implementations

DeltaReader

CsvFilterReader

GoldWriter

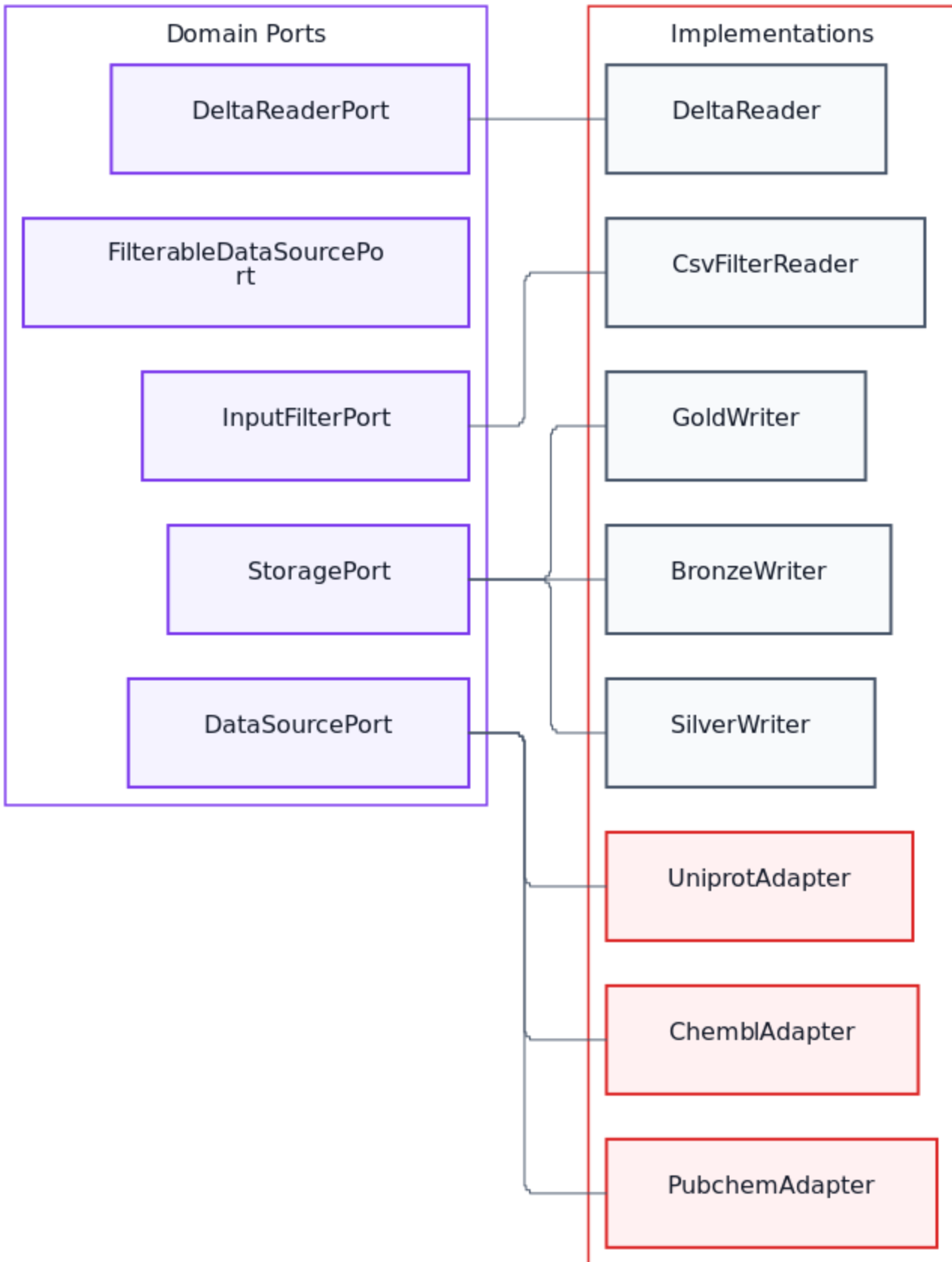
BronzeWriter

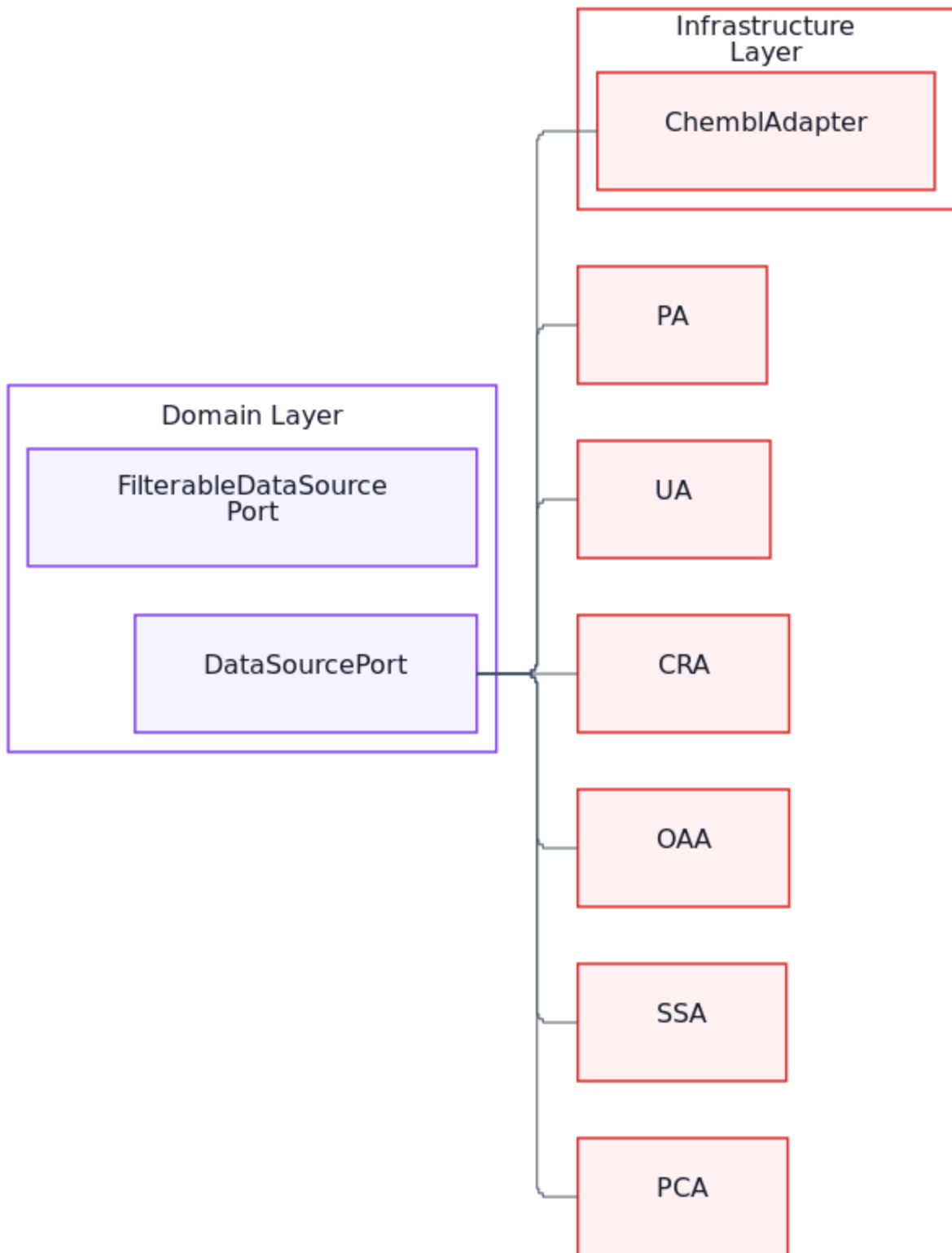
SilverWriter

UniprotAdapter

ChEMBLAdapter

PubchemAdapter





Domain Ports

LoggerPort

LockPort

CircuitBreakerPort

MemoryMonitorPort

CheckpointPort

MetricsPort

TracingPort

ShutdownPort

Implementations

UnifiedLogger

MemoryLock

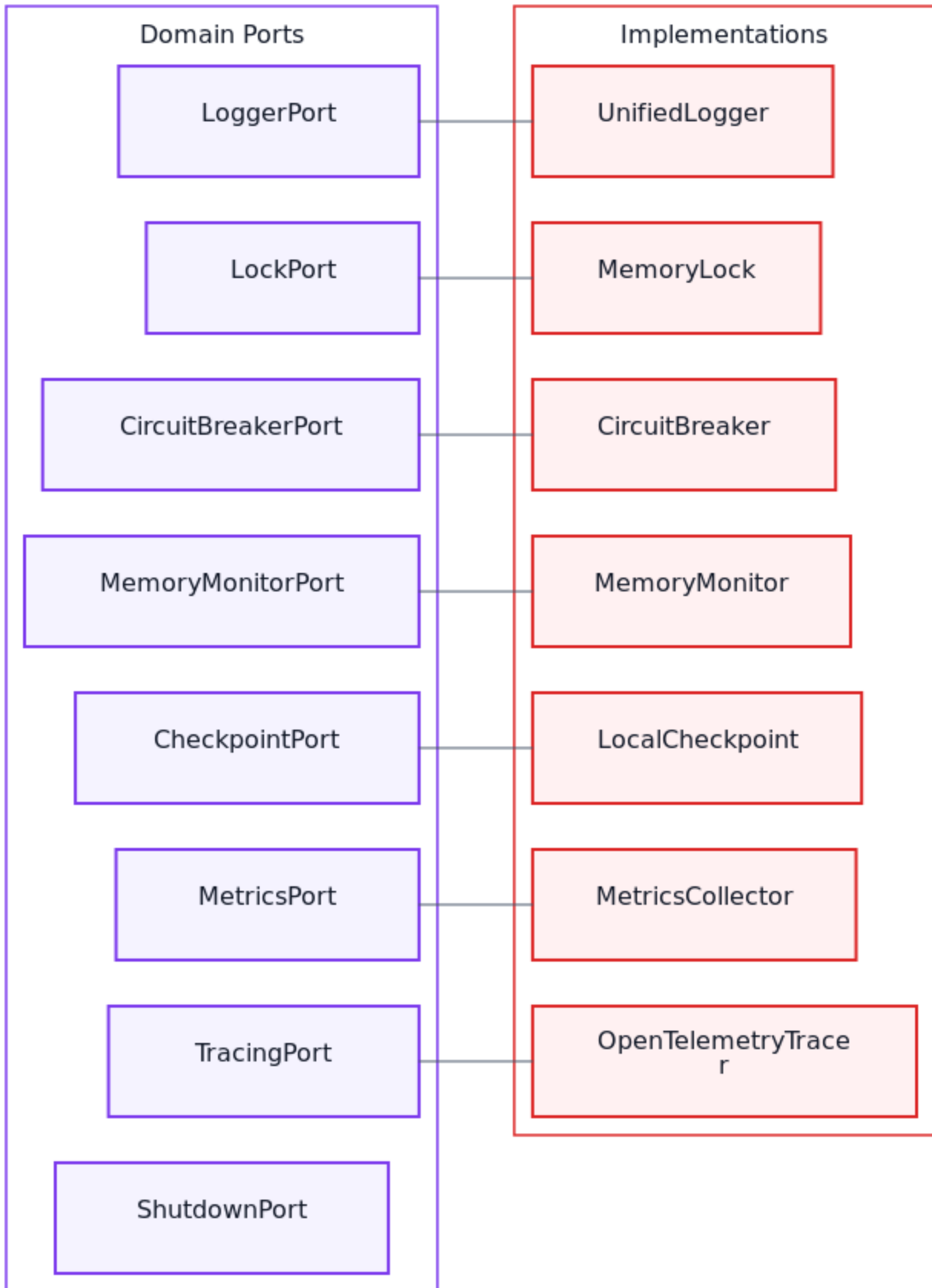
CircuitBreaker

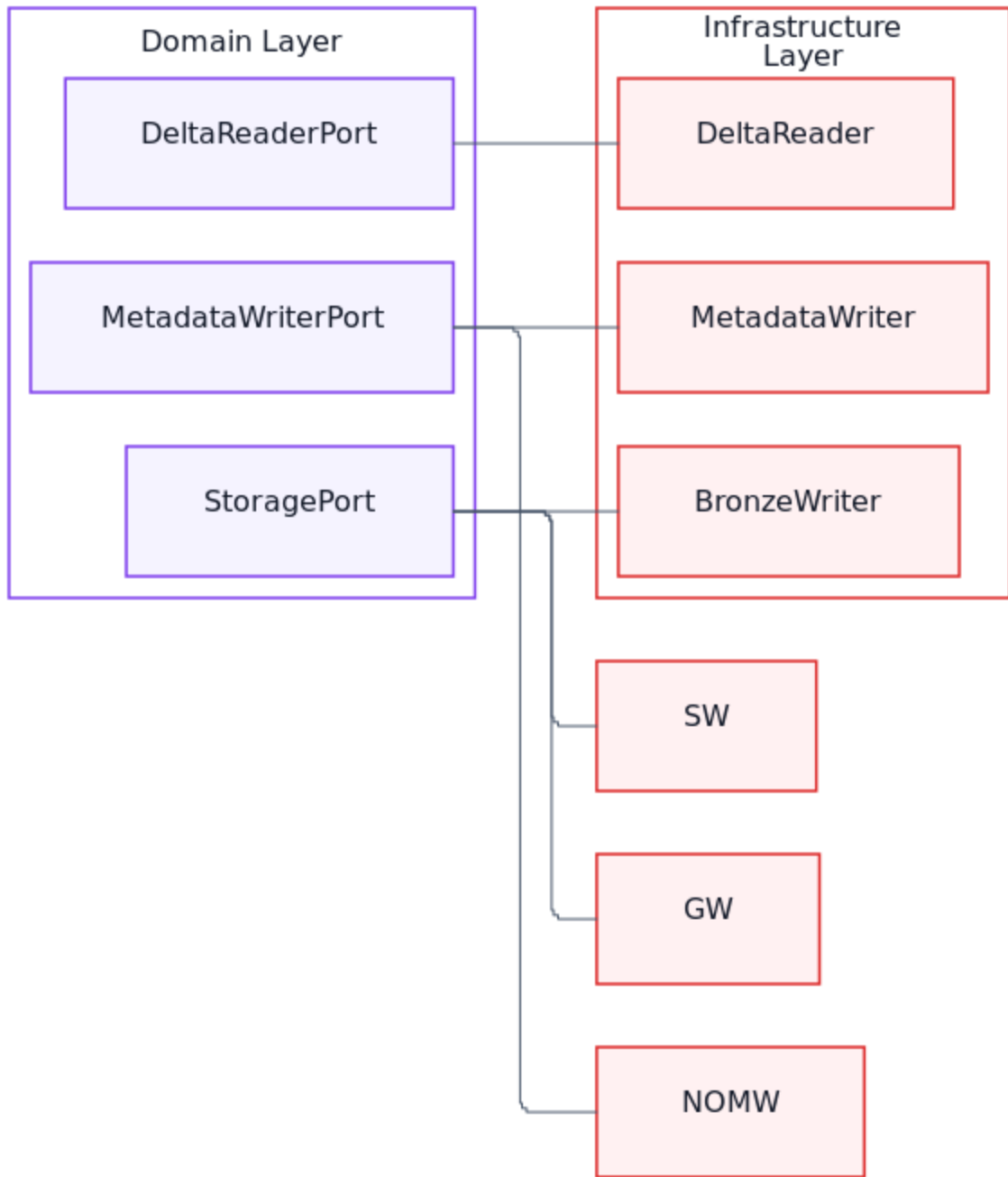
MemoryMonitor

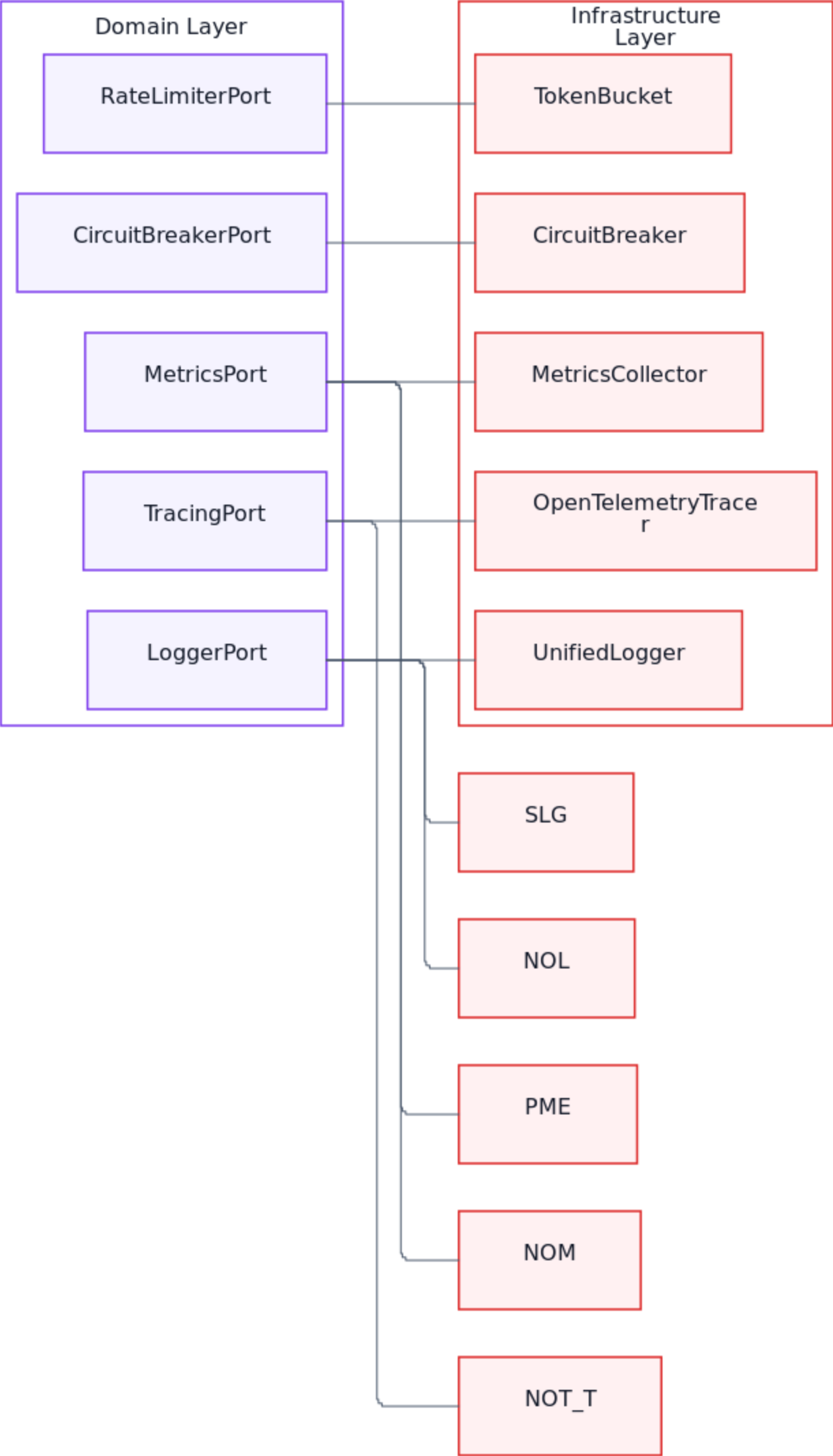
LocalCheckpoint

MetricsCollector

OpenTelemetryTrace
r







Domain Ports

GoldDQAnalyzerPort

BronzeDQAnalyzerPort

SilverDQAnalyzerPort

DQReportWriterPort

SilverValidatorPort

GoldValidatorPort

QuarantinePort

Application

GoldDQAnalyzer

BronzeDQAnalyzer

SilverDQAnalyzer

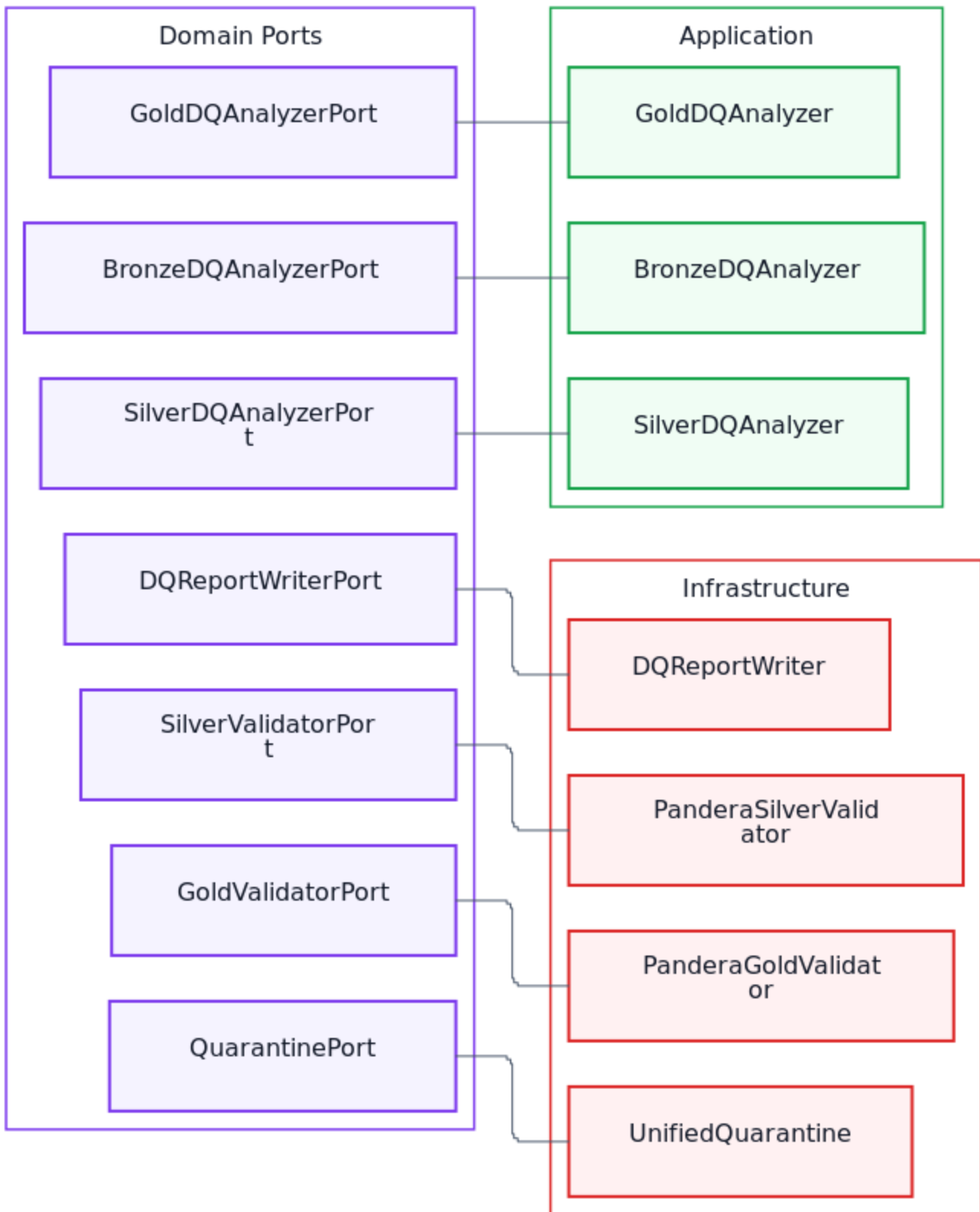
Infrastructure

DQReportWriter

PanderaSilverValidator

PanderaGoldValidator

UnifiedQuarantine





Domain Operational Ports

LockPort

CheckpointPort

LoggerPort

MetricsPort

TracingPort

CircuitBreakerPort

MemoryMonitorPort

ShutdownPort

Infrastructure Implementations

MemoryLock

LocalCheckpoint

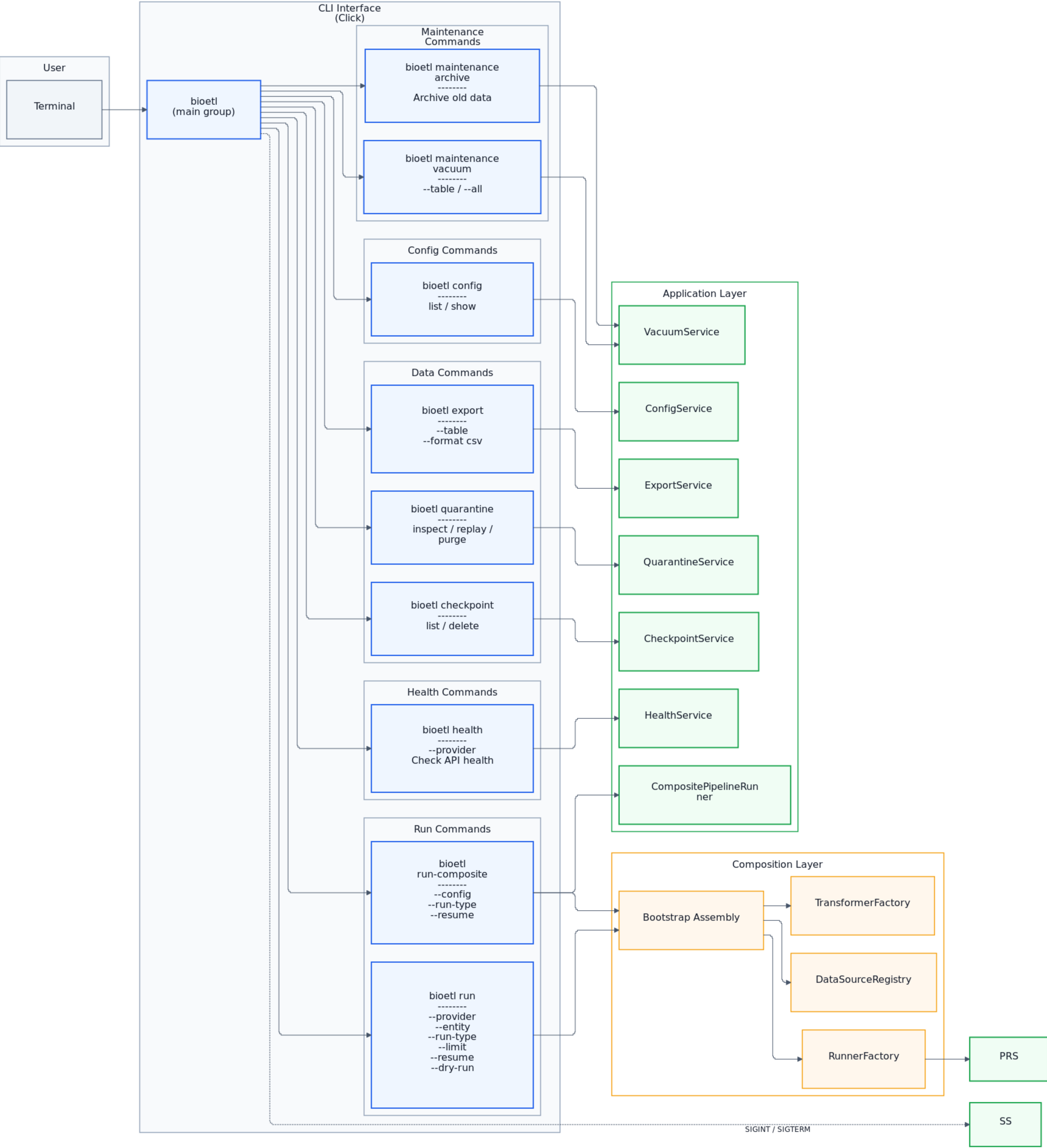
UnifiedLogger

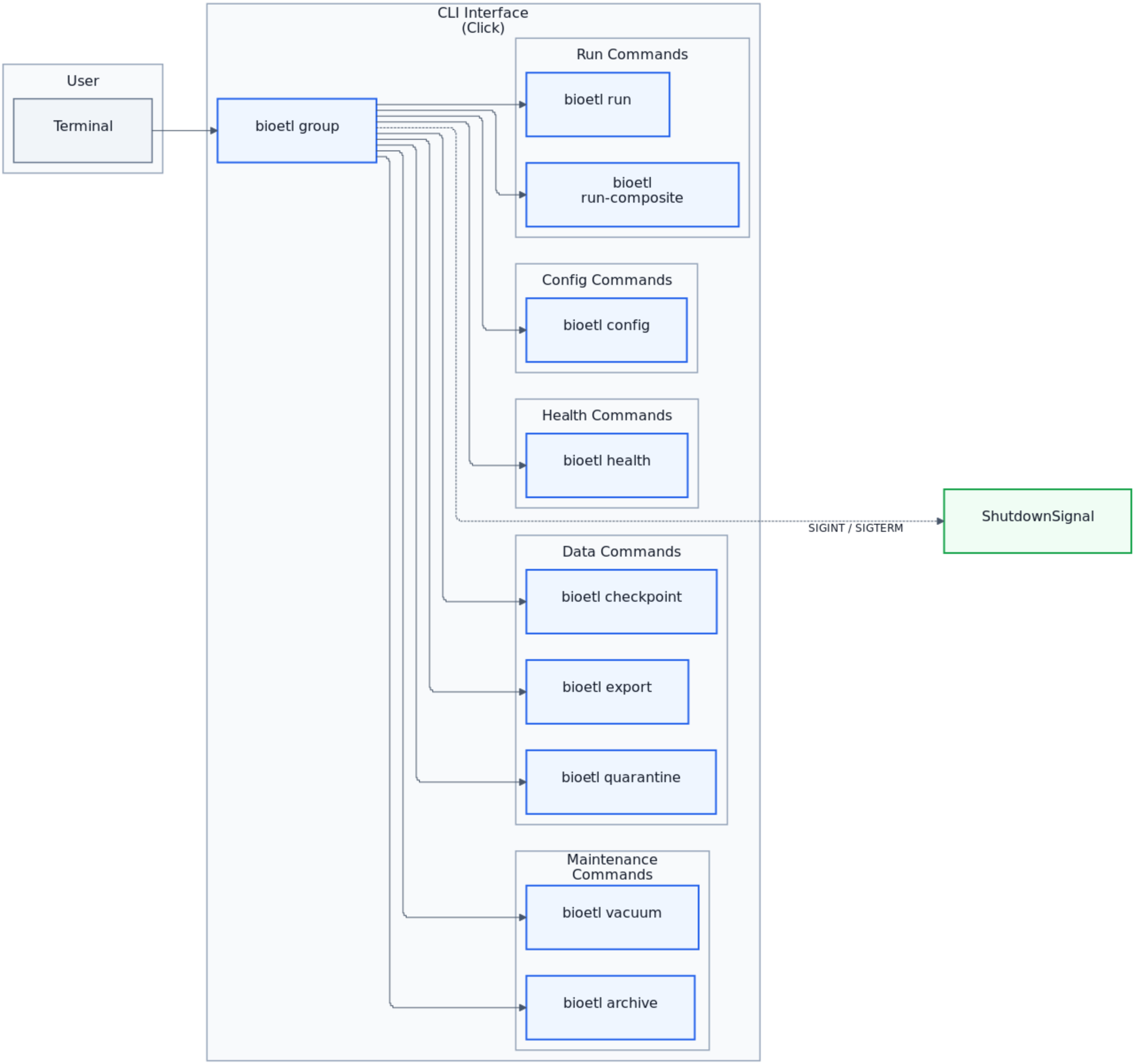
MetricsCollector

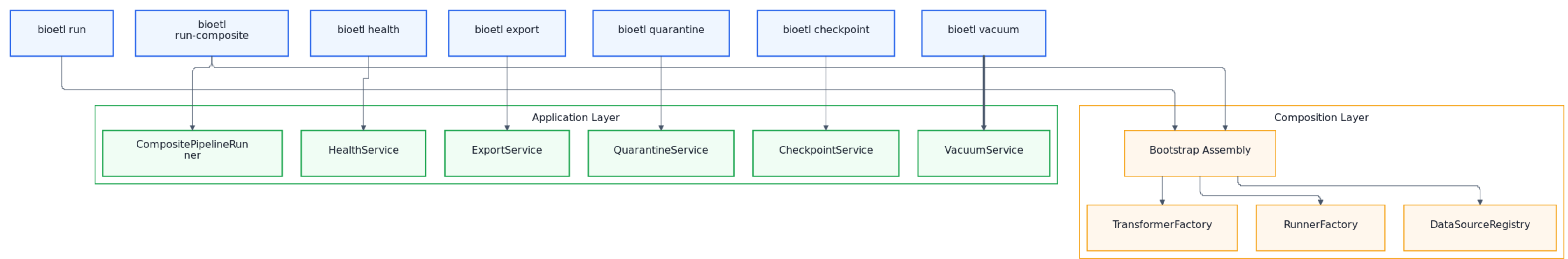
OpenTelemetryTrace
r

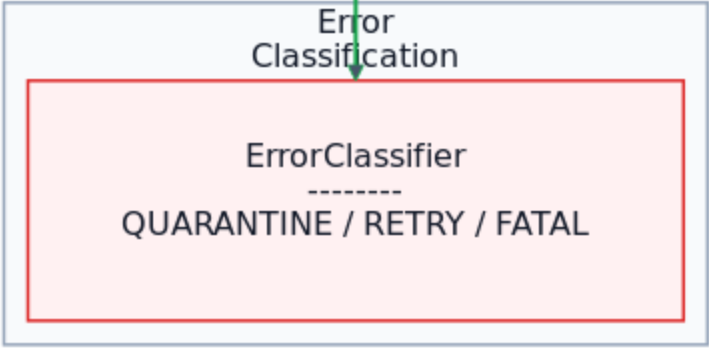
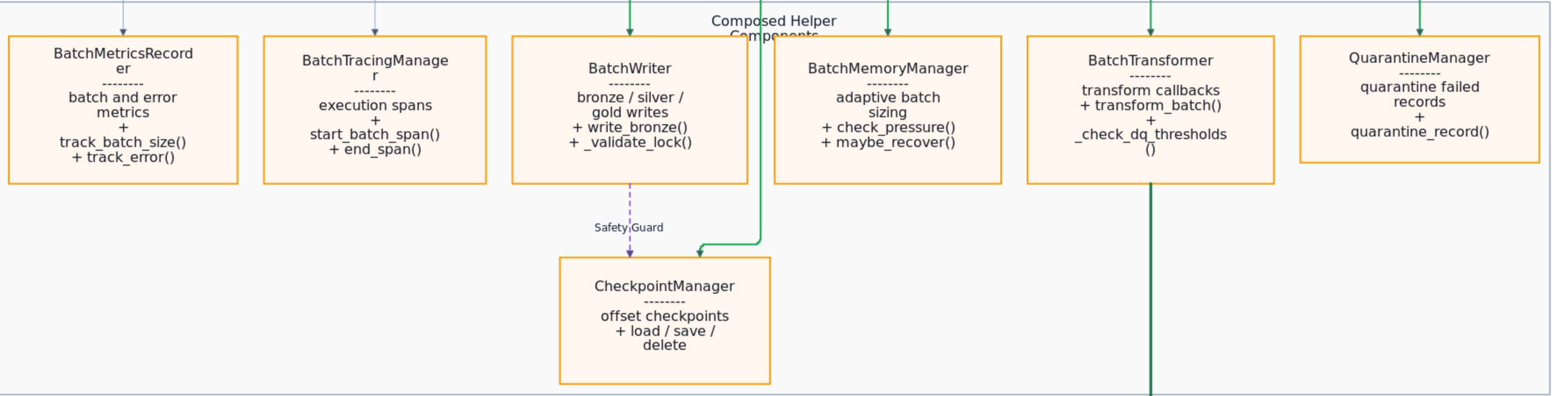
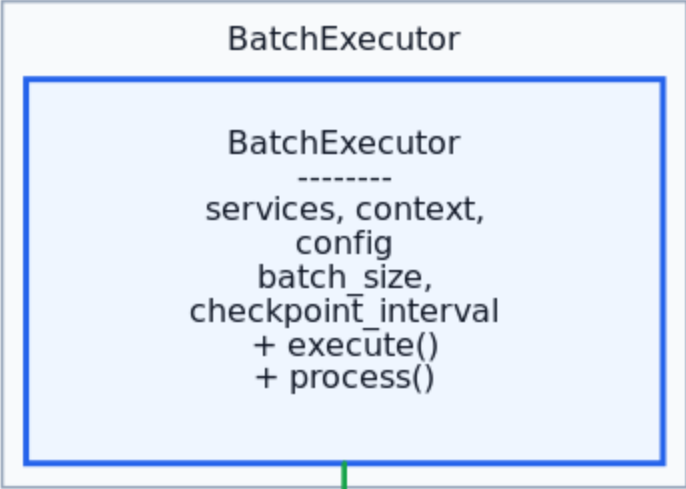
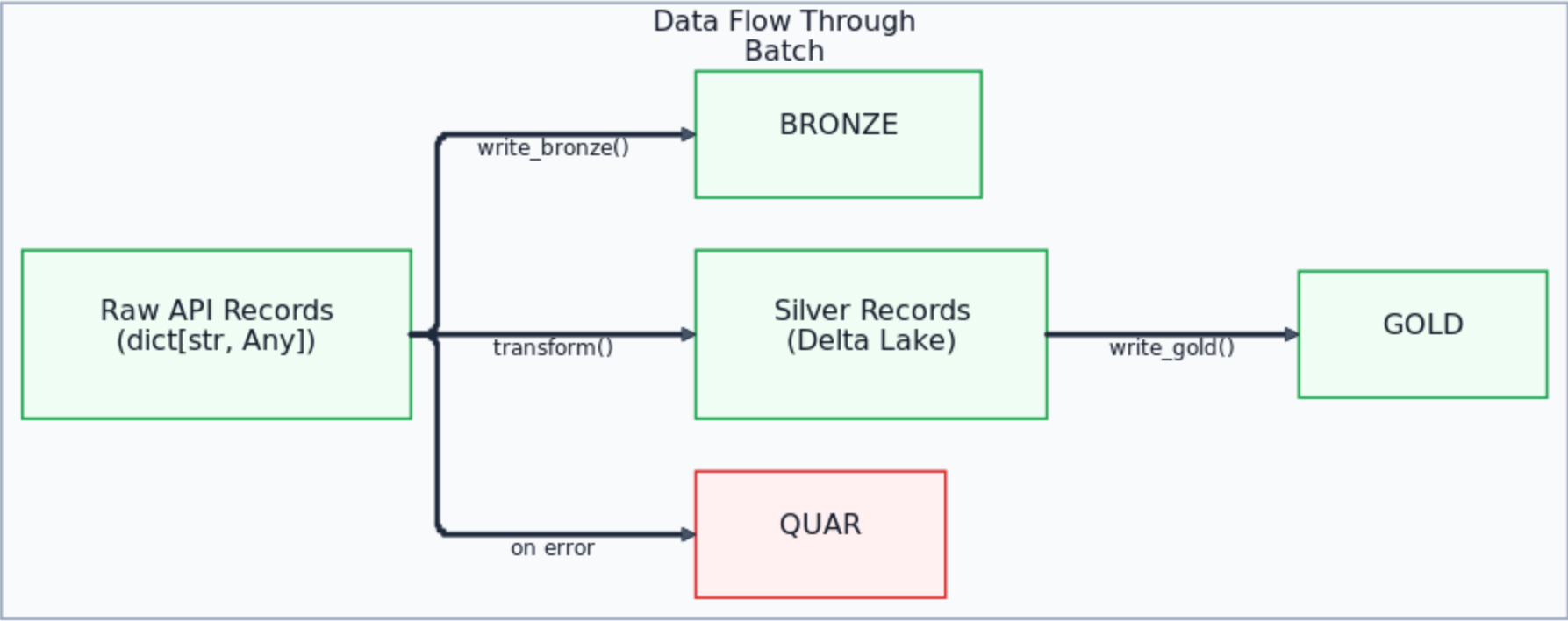
CircuitBreaker

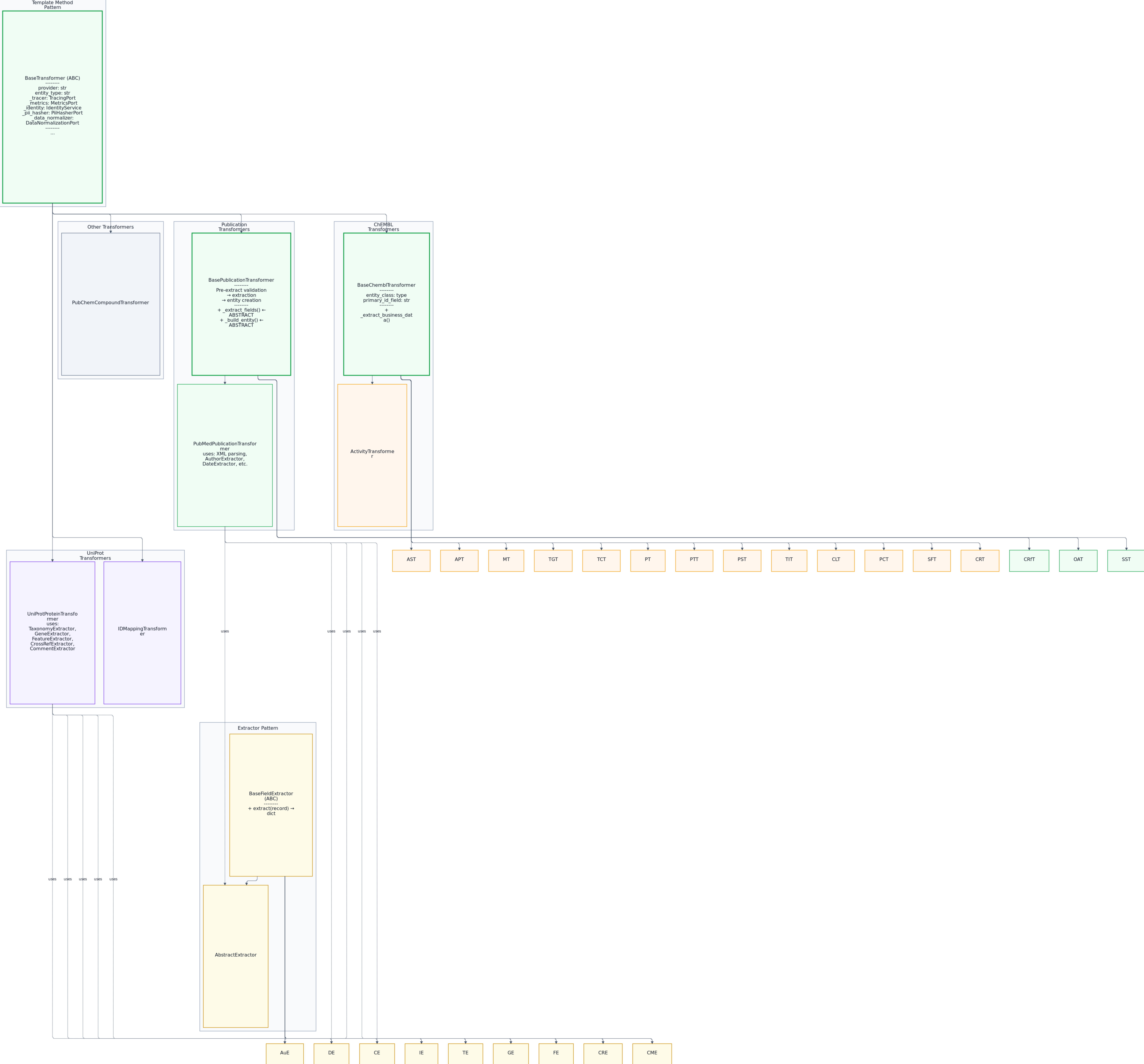
MemoryMonitor

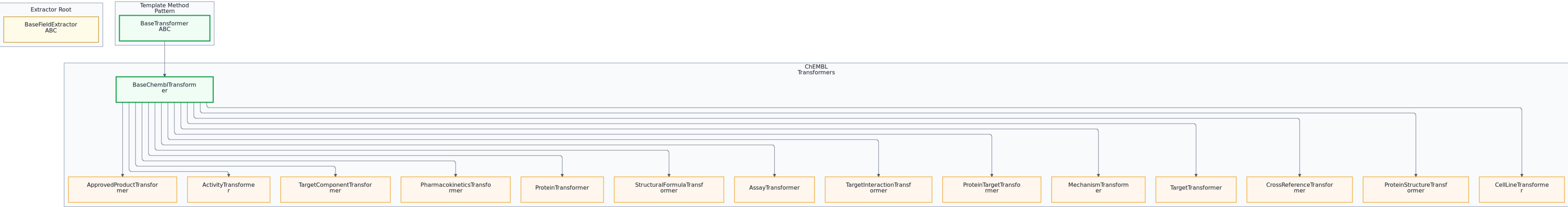


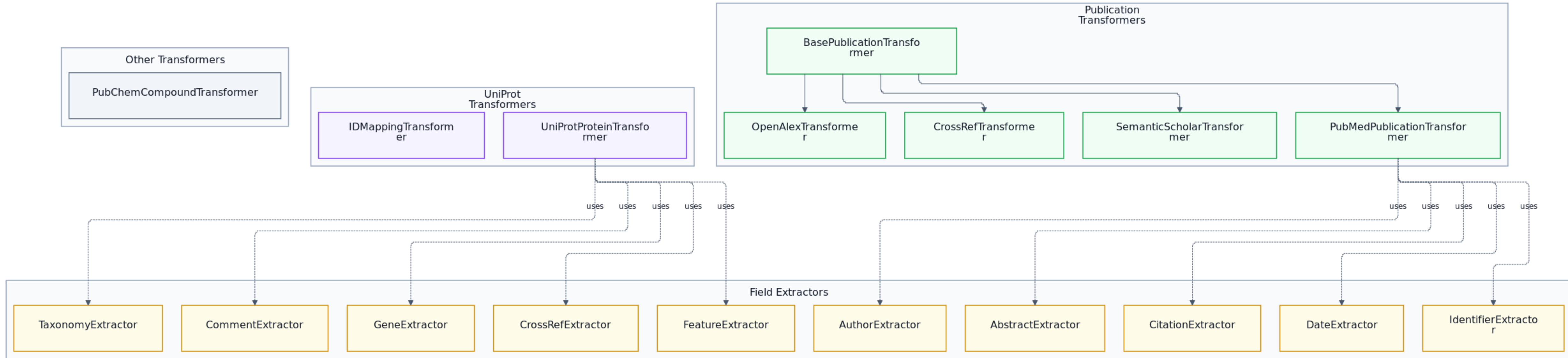


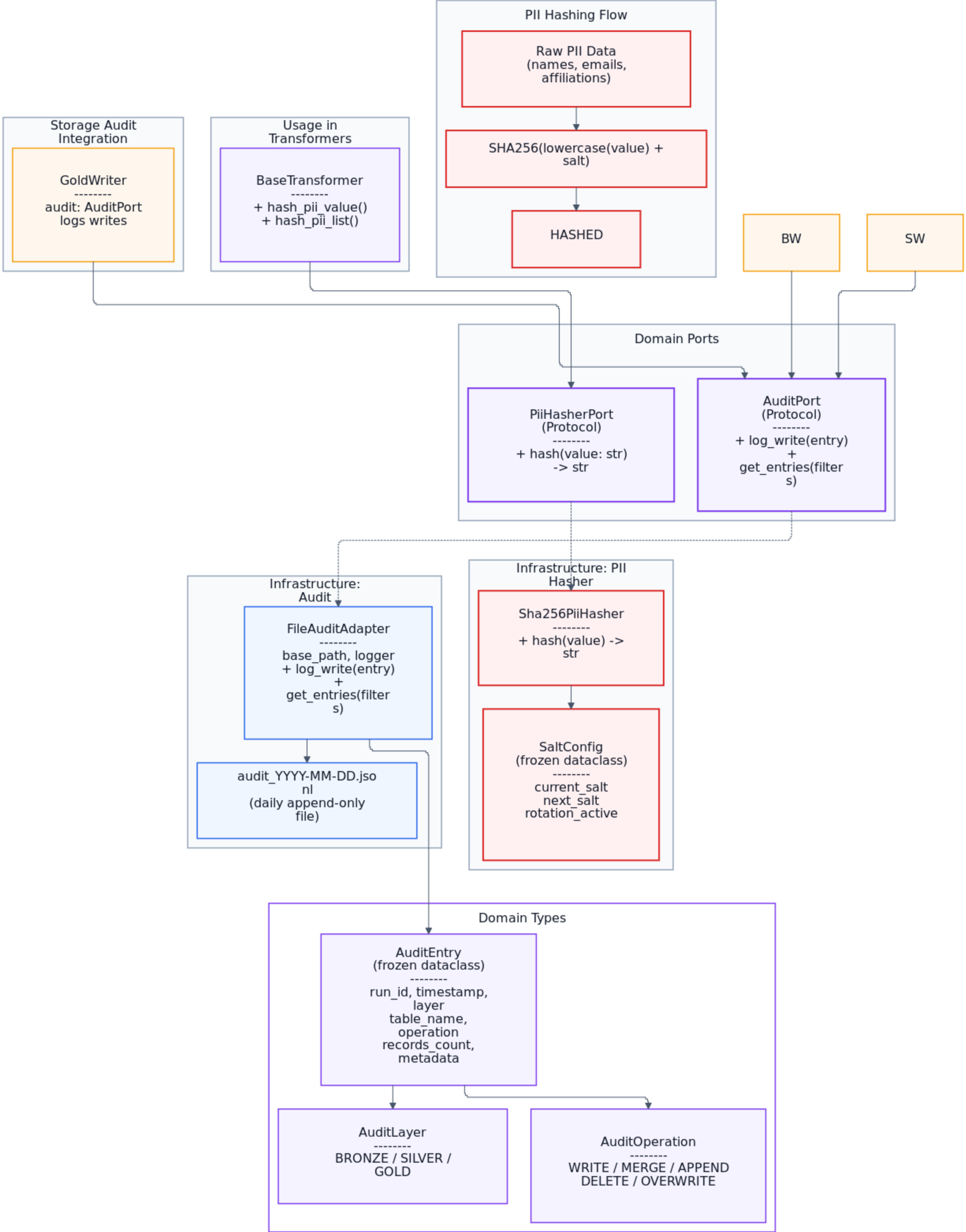


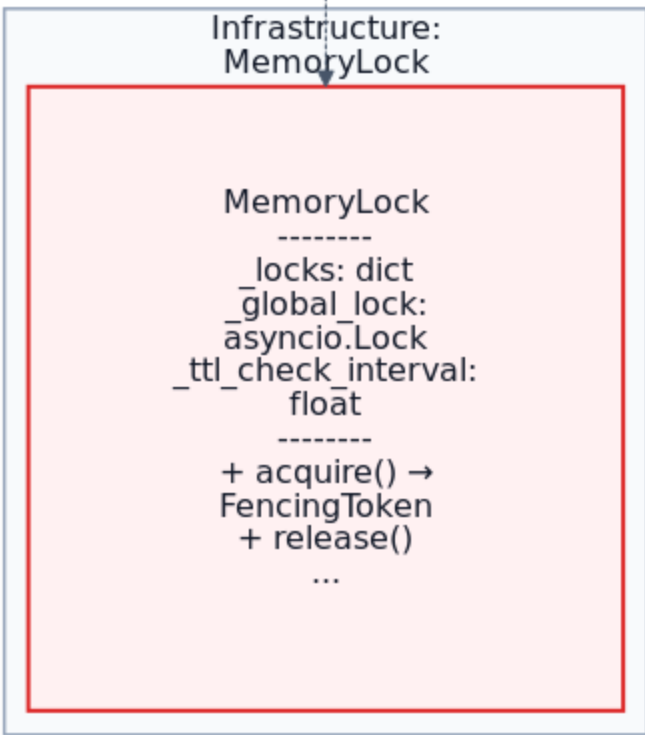
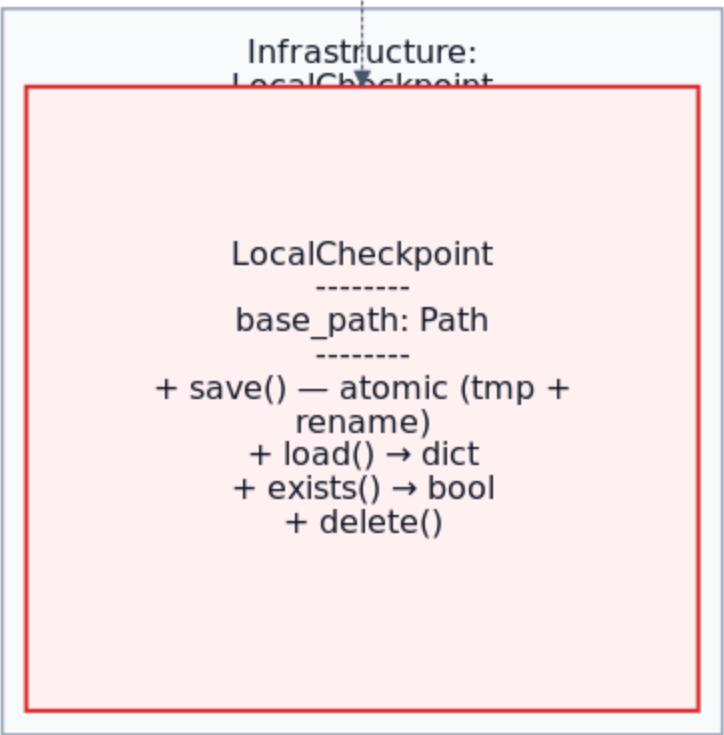
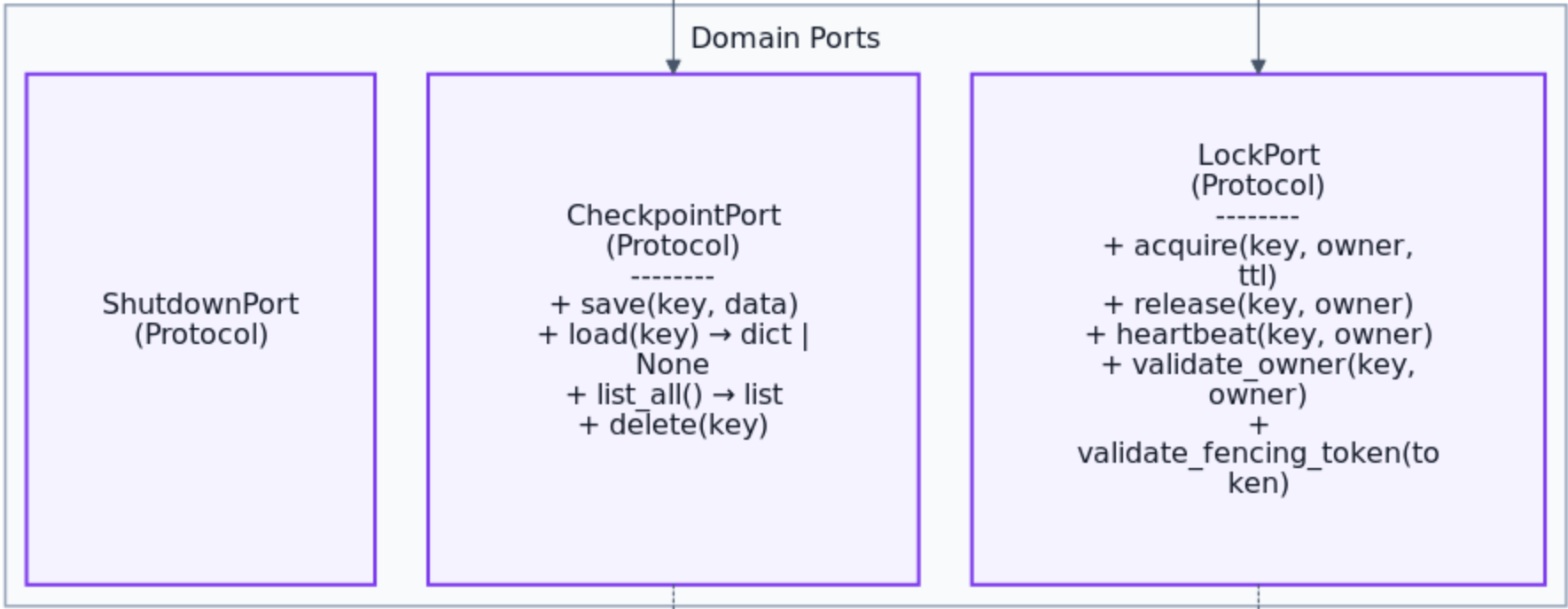
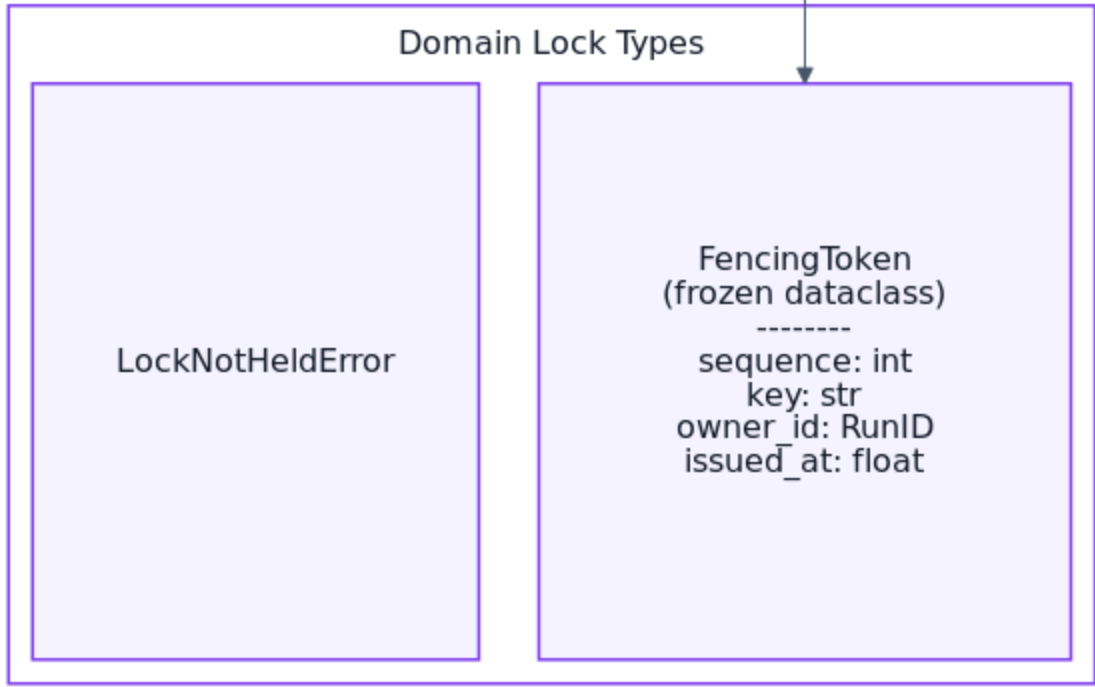
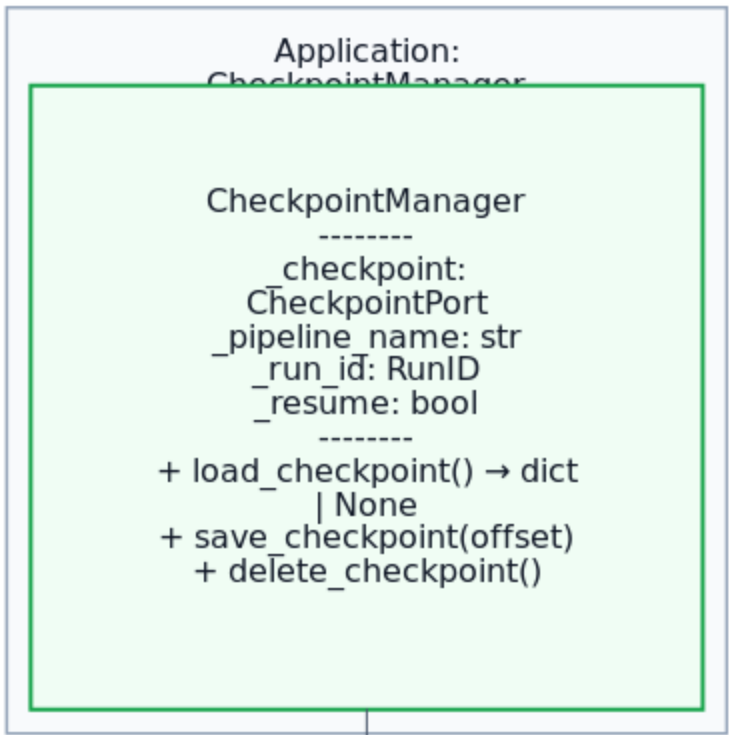
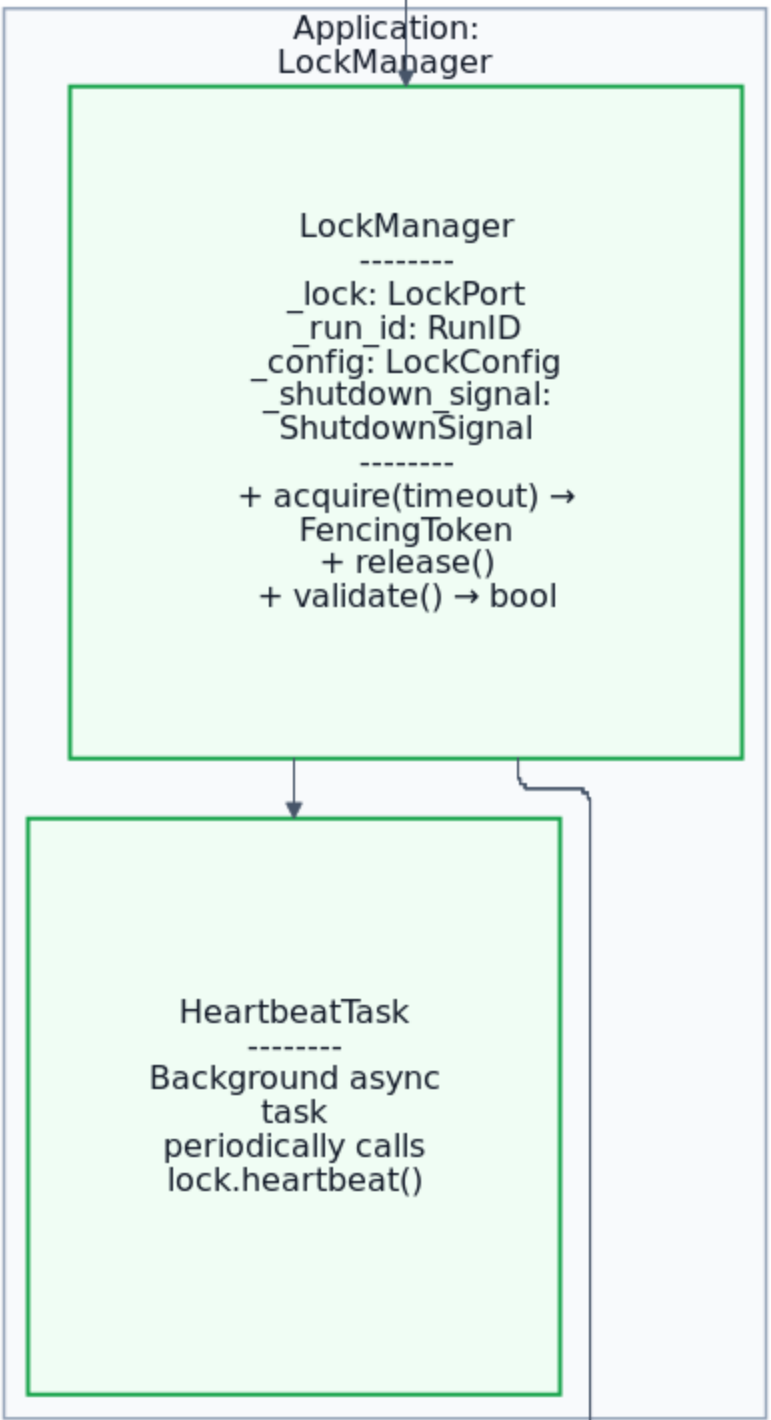
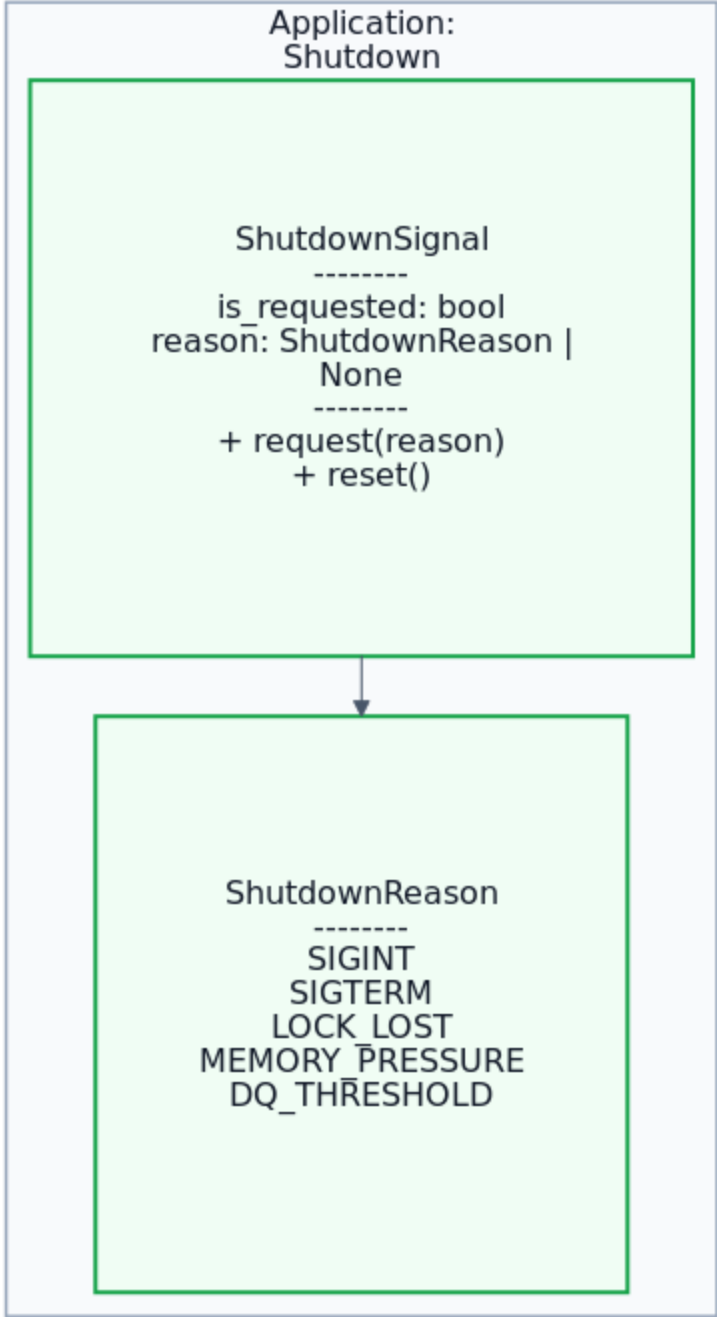
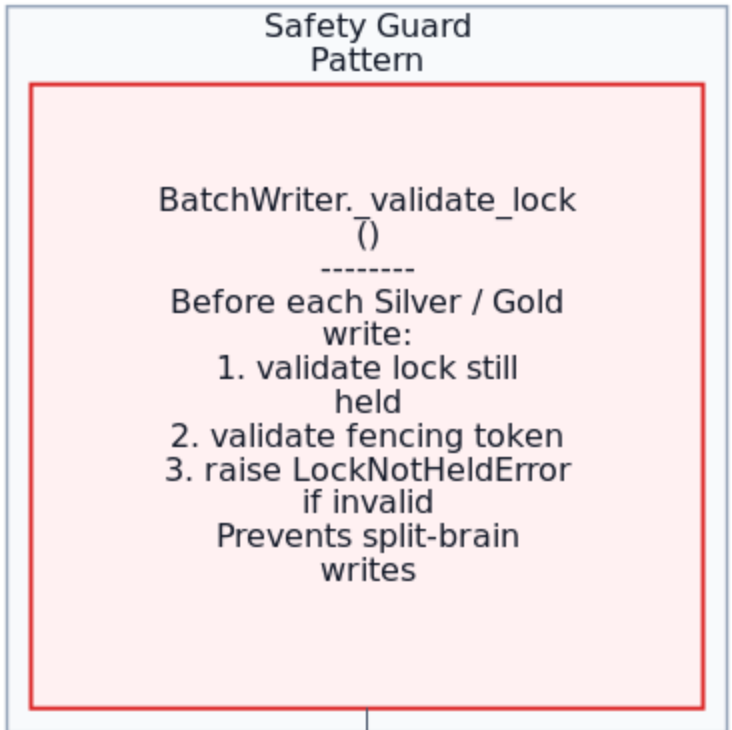
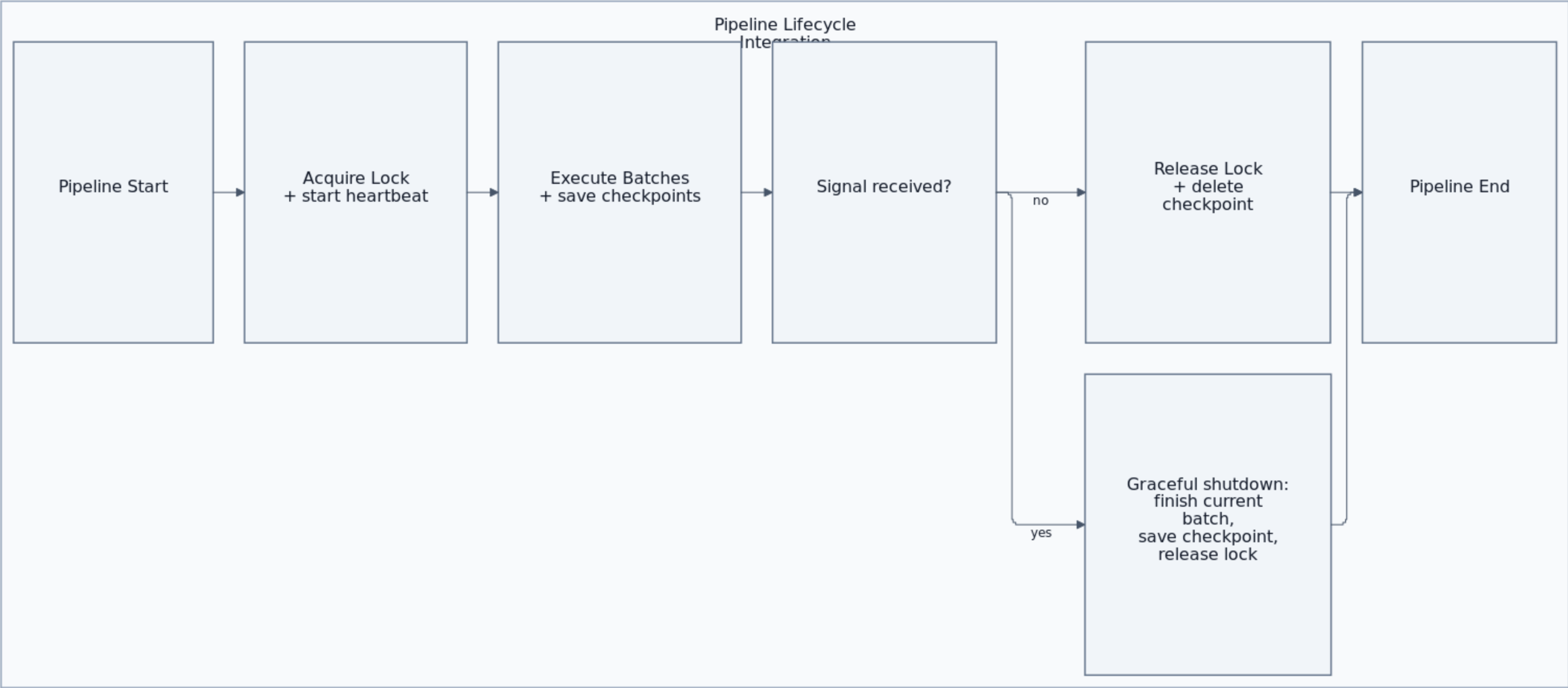


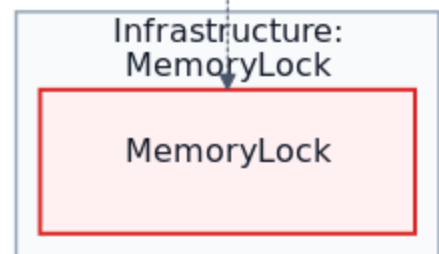
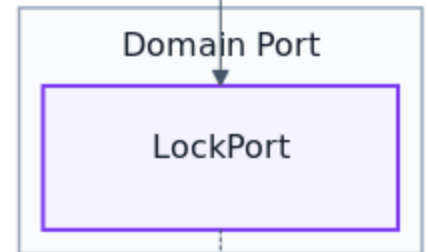
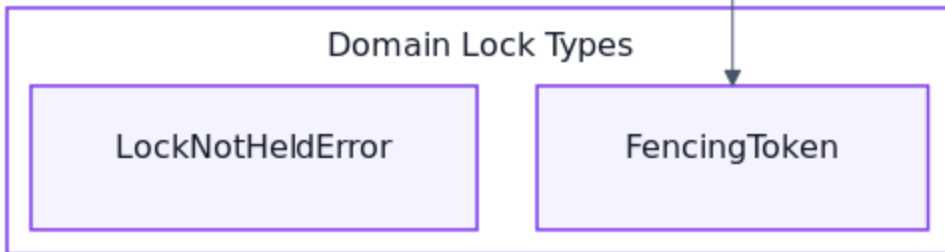
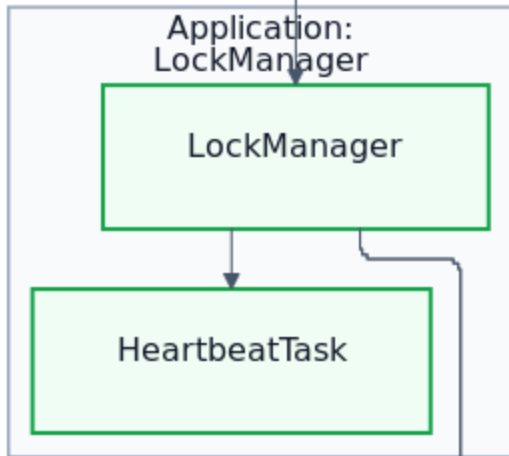
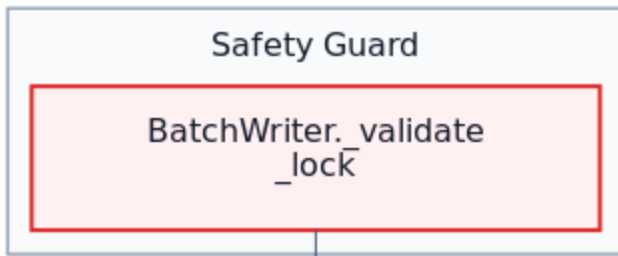












Pipeline Lifecycle

Pipeline Start

Acquire Lock

Execute Batches

Signal received?

no

Release Lock

Pipeline End

yes

Graceful Shutdown

Application:
Shutdown

ShutdownSignal

ShutdownReason

Application:
CheckpointManager

CheckpointManager

Domain Ports

ShutdownPort

CheckpointPort

Infrastructure:
LocalCheckpoint

LocalCheckpoint